

A POSTERIORI ERROR ANALYSIS OF FINITE ELEMENT METHOD FOR LINEAR NONLOCAL DIFFUSION AND PERIDYNAMIC MODELS

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ABSTRACT. In this paper, we present some results on a posteriori error analysis of finite element methods for solving linear nonlocal diffusion and bond-based peridynamic models. In particular, we aim to propose a general abstract framework for a posteriori error analysis of the peridynamic problems. A posteriori error estimators are consequently prompted, and the connection between non-local a posteriori error estimation and classical local estimation is studied. Numerical experiments are also given to test the theoretical conclusions.

1. INTRODUCTION

Since peridynamic (PD) models were first introduced by Silling in [33], their effectiveness has been demonstrated in a number of applications, e.g. fracture and failure of composites, fracture of polycrystals, nanofiber networks and so on, where numerical simulations of singular behavior such as crack growth or damage are involved. Peridynamics is becoming an attractive emerging tool for the multiscale material simulations. On one hand, it is an integral-type nonlocal continuum theory which provides a more general set-up than the classical partial differential equation (PDE) based elasticity theories for physical problems with discontinuities. On the other hand, it can also be viewed as a continuum version of molecular dynamics.

Along with research efforts on the modeling and application to practical problems, there have been some recent studies concerning the mathematical and numerical analysis of PD models. For example, some wellposedness studies of Cauchy problems for linear PD models have been given in [4, 23, 24, 25, 26, 27]. Similar studies for initial and nonlocal boundary value problems have been considered in [39] which has also investigated the issue of convergence of finite dimensional approximations of the linear bond-based PD models. There have been a variety of numerical methods implemented for the approximations of PD models including finite difference, finite element, quadrature and particle based methods [2, 6, 14, 19, 31, 35]. Convergence analysis and conditioning estimates for the discretized system have been given in [2, 3, 22, 39]. Moreover, connections between linear PD models and the continuum elasticity models have been established in [36, 25, 24, 39] as the ratio

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of the PD horizon to the effective length scale decreases. In [28], a nonlocal vector calculus was developed for the study of boundary value problems of scalar nonlocal diffusion equations. Extensions to vector and tensor fields and applications to nonlocal systems of variational equations and abstract nonlocal balance laws including the peridynamic models are presented in [21]. The framework presented in [21] has been utilized in recent mathematical and numerical analysis of nonlocal variational problems [22, 23]. Our current study is a part of the ongoing efforts.

In this work, we continue the works started in [39] and [22] on the numerical analysis of finite element method for bond-based linear steady state PD models and the associated nonlocal diffusion equations. Motivated by the need of adaptive computation for large scale simulations based on the nonlocal models with possible singular solutions, we mainly focus, in this work, on the development of a posteriori error estimators for nonlocal model equations to provide more theoretical guidance on the design of automatic error control and adaptive numerical schemes. Historically, a posteriori error estimations have played a very important role in adaptive mesh generation and algorithm design for numerical PDEs. Theoretical and systematical study of a posteriori error estimators for finite element approximation began in the late 1970's [11], and since then there have been a lot of studies in the literature, see for example, [1, 8, 9, 10, 12, 17, 20, 29, 37] and the references cited therein. There have also been many works on adaptive methods and a posteriori analysis for integral equations, see [16, 18, 32]. While adaptive refinement and coarsening method for PD model have been studied in [13, 14, 34, 38], there has not been any serious attempt to derive rigorous a posteriori error estimators. On the other hand, although PD models are integral equations, given the variational formulations presented in [21] based on the framework of nonlocal calculus and nonlocal balance laws, there is a close resemblance between the nonlocal PD models and the classical differential equations, which can be utilized to develop a posteriori error estimates for PD models. The latter is the main objectives of the current work. We provide a rigorous a posteriori error estimator which is shown to be both reliable and efficient. Moreover, we pay close attention to not only the connections but also the key differences between the a posteriori analysis of local and nonlocal models. Dependence of the error estimators on the model parameters is also considered. Some preliminary numerical experiments provide further substantiation of our theoretical analysis.

The paper is organized as follows. To prepare for the necessary background, we present in Section 2 some definitions of the nonlocal operators as constructed in [21] and describe the basic model equations. We also briefly recall the key features of the a posteriori error analysis for PDE models so as to offer contrast to the framework developed here for the nonlocal models. In order to present the essential features of the analysis without too much technical complications, we begin our analysis in a simpler setting, namely, we first consider a linear scalar-valued nonlocal diffusion model in Section 3. An a posteriori error estimator for such a model is introduced and its local limit is analyzed. Then in Section 4, we move on to develop a posteriori estimators for bond-based linear PD models in an analogous manner but with a focus on treating the extra complication due to the system nature. We present several numerical experiments in Section 5 to test our theories, and some conclusions are given in Section 6.

2. NONLOCAL OPERATORS AND MODEL EQUATIONS

Let us first recall some nonlocal operators as introduced in [21]. Let Ω be a bounded, open, connected spatial domain in $\mathbb{R}^d$ and

$$(2.1) \quad \Omega = (\Omega_s \cup \Omega_c) \cup (\bar{\Omega}_s \cap \bar{\Omega}_c)^o,$$

where Ω_s denotes the *solution domain*, Ω_c denotes the *constraint domain* such that $\Omega_s \cap \Omega_c = \emptyset$ and $(\bar{\Omega}_s \cap \bar{\Omega}_c)^o$ denotes the interior of the region $\bar{\Omega}_s \cap \bar{\Omega}_c$. We assume that Ω_s has convex polygonal boundaries with Ω_c being a domain surrounding Ω_s .

Let $\beta(\mathbf{x}, \mathbf{x}') : \Omega \times \Omega \rightarrow \mathbb{R}$ be a scalar non-negative two-point function which is symmetric, that is, $\beta(\mathbf{x}, \mathbf{x}') = \beta(\mathbf{x}', \mathbf{x})$, and has a compact support such that

$$\beta(\mathbf{x}, \mathbf{x}') = 0 \quad \forall \mathbf{x}, \mathbf{x}' \in \Omega, \quad \text{with } |\mathbf{x}' - \mathbf{x}| > \delta$$

for some constant number $\delta > 0$. We also assume it is non-degenerate in the sense that there exists a constant $\beta_0(\delta) > 0$ such that

$$\beta(\mathbf{x}, \mathbf{x}') \geq \beta_0(\delta) \quad \forall \mathbf{x}, \mathbf{x}' \in \Omega, \quad \text{with } |\mathbf{x}' - \mathbf{x}| < \delta/2.$$

Here, $\delta/2$ may be replaced by other suitable constants smaller than δ to ensure the non-degeneracy of $\beta(\mathbf{x}, \mathbf{x}')$. For convenience of reference, we call δ the horizon parameter, following the naming convention given in [33].

Let $\alpha(\mathbf{x}, \mathbf{x}') : \Omega \times \Omega \rightarrow \mathbb{R}^d$ be an skew-symmetric two-point mapping such that

$$\alpha(\mathbf{x}', \mathbf{x}) + \alpha(\mathbf{x}, \mathbf{x}') = 0.$$

Given a two-point function $\mathbf{v} : \Omega \times \Omega \rightarrow \mathbb{R}^d$ and a one-point function $u : \Omega \rightarrow \mathbb{R}$, the nonlocal point divergence operator $\mathcal{D} : \Omega \rightarrow \mathbb{R}$, is defined as, see [21]

$$(2.2) \quad \mathcal{D}(\mathbf{v})(\mathbf{x}) := - \int_{\Omega} (\mathbf{v}(\mathbf{x}, \mathbf{x}') + \mathbf{v}(\mathbf{x}', \mathbf{x})) \cdot \alpha(\mathbf{x}, \mathbf{x}') d\mathbf{x}', \quad \forall \mathbf{x} \in \Omega,$$

and the adjoint operator $\mathcal{D}^*(u) : \Omega \times \Omega \rightarrow \mathbb{R}^d$ of $\mathcal{D}$ is given by

$$(2.3) \quad \mathcal{D}^*(u) := (u(\mathbf{x}') - u(\mathbf{x}))\alpha(\mathbf{x}, \mathbf{x}'), \quad \forall \mathbf{x}', \mathbf{x} \in \Omega.$$

2.1. Nonlocal diffusion equations. In section 3, we consider the *nonlocal constrained-value problem*

$$(2.4) \quad \begin{cases} \mathcal{D}(\beta \mathcal{D}^*(u(\mathbf{x}))) = f(\mathbf{x}) & \text{for } \mathbf{x} \in \Omega_s \\ u(\mathbf{x}) = 0 & \text{for } \mathbf{x} \in \Omega_c, \end{cases}$$

where the first equation is called a nonlocal diffusion equation, while the second equation is a *nonlocal homogeneous Dirichlet constraint* (nonlocal boundary condition). The above problem has a natural variational characterization [22, 23] and the natural function space associated with the solution of (2.4) is given by

$$(2.5) \quad V_c := \{u \in L^2(\Omega) : \int_{\Omega} \int_{\Omega} \beta(\mathbf{x}, \mathbf{x}') |\mathcal{D}^*(u)(\mathbf{x}, \mathbf{x}')|^2 d\mathbf{x}' d\mathbf{x} < \infty, u(\mathbf{x}) = 0 \forall \mathbf{x} \in \Omega_c\}.$$

More precise characterizations of V_c are given in later sections.

2.2. Nonlocal bond-based linear peridynamic models. While the scalar non-local diffusion equations have many applications on its own [15], the vectorized form of (2.4) can have the same form as that of a linear bond-based peridynamic model. The latter may be more conveniently expressed with some additional operators in the nonlocal vector calculus framework. Indeed, given the scalar two point mapping $v : \Omega \times \Omega \rightarrow \mathbb{R}$, the *nonlocal point gradient operator* $\mathcal{G}(v) : \Omega \rightarrow \mathbb{R}^d$ is defined as

$$(2.6) \quad \mathcal{G}(v)(\mathbf{x}) := - \int_{\Omega} (v(\mathbf{x}, \mathbf{x}') + v(\mathbf{x}', \mathbf{x})) \boldsymbol{\alpha}(\mathbf{x}, \mathbf{x}') d\mathbf{x}', \quad \forall \mathbf{x} \in \Omega.$$

Given the point mapping $\mathbf{u}(\mathbf{x}) : \Omega \rightarrow \mathbb{R}^d$, the adjoint operator $\mathcal{G}^*(\mathbf{u}) : \Omega \times \Omega \rightarrow \mathbb{R}$ corresponding to the point operator $\mathcal{G}$ defined above is given by

$$(2.7) \quad \mathcal{G}^*(\mathbf{u})(\mathbf{x}, \mathbf{x}') = (\mathbf{u}(\mathbf{x}') - \mathbf{u}(\mathbf{x})) \cdot \boldsymbol{\alpha}(\mathbf{x}, \mathbf{x}'), \quad \forall \mathbf{x}, \mathbf{x}' \in \Omega.$$

Following the discussions in [21, 23], given the $\beta(\mathbf{x}, \mathbf{x}')$ and $\boldsymbol{\alpha}$ defined above, we may have the following *nonlocal constrained-value problem* for the bond-based linear peridynamic model

$$(2.8) \quad \begin{cases} \mathcal{G}(\beta \mathcal{G}^*(\mathbf{u}(\mathbf{x}))) = \mathbf{f}(\mathbf{x}) & \text{for } \mathbf{x} \in \Omega_s \\ \mathbf{u}(\mathbf{x}) = 0 & \text{for } \mathbf{x} \in \Omega_c. \end{cases}$$

2.3. More discussions on the model problems. For both the model equations (2.4) and (2.8), it is straightforward to write down their variational form using the nonlocal operators and the nonlocal balance laws [22, 23]. This in turn allows the construction of Galerkin approximations using, for instance, finite element spaces.

For these model equations, the properties of solutions of the model equations depend crucially on the *kernel function* $\gamma = \boldsymbol{\alpha} \cdot \boldsymbol{\alpha} \beta$. In the literature, a type of kernel functions that have been most-widely studied is the class of non-negative integrable kernels.

Here, we make a simplifying assumption that the kernel function $\gamma : \Omega \times \Omega \rightarrow \mathbb{R}$ is a function which only depends on $\mathbf{z} = \mathbf{x}' - \mathbf{x}$. Moreover, the integrable kernel function is assumed to satisfy the property that there exist positive constants $c(\delta)$ and $M(\delta)$ such that

$$(2.9) \quad \begin{aligned} 0 < c(\delta) &\leq \int_{B_\delta(0)} \gamma(\mathbf{z}) d\mathbf{z} \leq M(\delta) < \infty, \\ \gamma(\mathbf{z}) &\geq 0 \quad \forall |\mathbf{z}| \leq \delta, \quad \gamma(\mathbf{z}) = 0 \quad \forall |\mathbf{z}| > \delta. \end{aligned}$$

More general and non-integrable kernels can also be considered [24]. The non-integrable case corresponds to a broader class of problems which, for instance in the peridynamic context, still retains well-defined elastic moduli and leads to smoothing effect of the solution operators. However, the case corresponding to (2.9) is of particular interests in many practical applications of the peridynamic modeling of materials. Moreover, for $f \in L^2(\Omega)$, given the assumption (2.9), one expects that the solutions of (2.4) and (2.8) are in $L^2(\Omega)$ and $(L^2(\Omega))^d$, respectively, so that the solution operators do not have any extra smoothing effect. This situation sets such nonlocal models apart from their local limit and is thus the main focus of the present study. We refer to [3, 28, 22, 5, 39] for additional discussions of the well-posedness for the scalar case of the model equations and [39, 23] on PD state-based and bond-based systems. To avoid some technical complications which are not important for practical applications, additional assumptions are made in later sections to simplify the analysis.

2.4. Finite element discretization and local a posteriori error analysis.

We now briefly recall a posteriori error estimate for PDE problems. We choose to focus on the residual-based a posteriori error estimate which has been discussed extensively in the literature, see, e.g. [37]. We take, for example, the following model equation:

$$(2.10) \quad \begin{cases} -\nabla \cdot (\mathbb{D}\nabla u(x)) = f(x) & \text{for } x \in \Omega_s \\ u(x) = 0 & \text{for } x \in \partial\Omega_s, \end{cases}$$

where we assume that $f \in L^2(\Omega_s)$ and $\mathbb{D}(\mathbf{x})$ is an $d \times d$ symmetric positive definite matrix for any $\mathbf{x} \in \Omega_s$. We use u to denote the exact solution of the PDE problem (2.10).

Now, let $\{K\}$ denote a spatial simplicial mesh of Ω_s ($\Omega_s = \cup K$) and $\mathbf{S}_{local}^h$ denote a finite dimensional subspace of $H_0^1(\Omega_s)$ defined with respect to the mesh $\{K\}$. That is, $\mathbf{S}_{local}^h$ denotes a space of continuous and piecewise smooth functions (polynomials) defined on $\{K\}$, and u^h is the Galerkin finite dimensional approximation of u in the subspace $\mathbf{S}_{local}^h$ defined by the following problem:

$$(2.11) \quad \begin{cases} \text{Find } u^h \in \mathbf{S}_{local}^h, \text{ such that for any } v^h \in \mathbf{S}_{local}^h, \\ \int_{\Omega_s} \mathbb{D}\nabla u^h \cdot \nabla v^h d\mathbf{x} = \int_{\Omega_s} f v^h d\mathbf{x}, \\ \text{and } u^h(\mathbf{x}) = 0 \text{ on } \partial\Omega_s. \end{cases}$$

To obtain a posteriori error estimate for (2.11), we apply integration by parts on each mesh element K which gives that for any $v \in H_0^1(\Omega_s)$,

$$(2.12) \quad \begin{aligned} & \int_{\Omega_s} \mathbb{D}\nabla(u - u^h) \cdot \nabla v d\mathbf{x} \\ &= \sum_K \left(\int_K (f + \nabla \cdot (\mathbb{D}\nabla u^h)) v d\mathbf{x} - \frac{1}{2} \int_{\partial K} [\![\mathbb{D}\nabla u^h]\!]_{\mathbf{r}} v ds \right), \end{aligned}$$

where σ is the set of all the interior edges (faces) of the mesh; $[\![\mathbb{D}\nabla u^h]\!]_{\mathbf{r}}$ denotes the normal jump of ∇u^h on an edge (face) $\mathbf{r}$ as defined by

$$[\![\mathbb{D}\nabla u^h]\!]_{\mathbf{r}} = \mathbb{D}\nabla u_K^h|_{\mathbf{r}} \cdot \bar{\mathbf{n}}_K + \mathbb{D}\nabla u_{K'}^h|_{\mathbf{r}} \cdot \bar{\mathbf{n}}_{K'},$$

where $\mathbf{r}$ is the common edge (face) of K and K' with outward normal $\bar{\mathbf{n}}_K$ and $\bar{\mathbf{n}}_{K'}$ respectively; u_K^h and $u_{K'}^h$ denote $u^h|_K$ and $u^h|_{K'}$ respectively.

In the classical residual-based a posteriori error analysis for (2.11), by making use of (2.12), the *interior residual* is defined by

$$(2.13) \quad r_*^h = f + \nabla \cdot (\mathbb{D}\nabla u^h) \quad \text{in } K,$$

and the *boundary residual* by

$$(2.14) \quad R_*^h = [\![\mathbb{D}\nabla u^h]\!]_{\mathbf{r}} \quad \text{on any interior edge(face) } \mathbf{r},$$

then, the *local a posteriori error estimator* (energy type) η_K associated with element K can be defined as

$$(2.15) \quad \eta_K^2 = h_K^2 \|r_*^h\|_{L^2(K)}^2 + h_K \|R_*^h\|_{L^2(\partial K)}^2,$$

where h_K denotes the size (mesh parameter) of K . The reliability and efficiency of the estimator have been proved in [11], i.e., the estimator neither underestimates nor overestimates the actual error $u - u^h$.

Next, we mimic the above framework to get a posteriori error approximation for some nonlocal linear equations. We note that our study also shows significant differences between a posteriori error analysis for the nonlocal and local models.

3. FINITE ELEMENT DISCRETIZATION OF LINEAR NONLOCAL DIFFUSION EQUATIONS

We now study the finite element discretization of (2.4).

3.1. A nonlocal bilinear form and the weak formulation. We first prompt the following lemmas

Lemma 3.1. *Assume the kernel function γ satisfies (2.9), then for any $u \in L^2(\Omega)$, we have*

$$(3.1) \quad \mathcal{D}(\beta \mathcal{D}^*(u)) \in L^2(\Omega).$$

Proof. For $u \in L^2(\Omega)$, we have by apply Cauchy-Schwartz inequality,

$$\begin{aligned} & \|\mathcal{D}(\beta \mathcal{D}^*(u))\|_{L^2(\Omega)}^2 \\ &= \int_{\Omega} \left| \int_{\Omega} 2\gamma(u(\mathbf{x}') - u(\mathbf{x})) d\mathbf{x}' \right|^2 d\mathbf{x} \\ &\leq 4 \int_{\Omega} \int_{\Omega} \gamma d\mathbf{x}' \int_{\Omega} \gamma(u(\mathbf{x}') - u(\mathbf{x}))^2 d\mathbf{x}' d\mathbf{x} \\ &= 4 \int_{\Omega} \gamma(\mathbf{z}) dz \int_{\Omega} \int_{\Omega} \gamma(u(\mathbf{x}') - u(\mathbf{x}))^2 d\mathbf{x}' d\mathbf{x} \\ &= 4 \int_{B_{\delta}(0)} \gamma(\mathbf{z}) d\mathbf{z} \int_{\Omega} \int_{\Omega} \gamma(u^2(\mathbf{x}') + u^2(\mathbf{x}) - 2u(\mathbf{x}')u(\mathbf{x})) d\mathbf{x}' d\mathbf{x} \\ (3.2) \quad &= 8 \int_{B_{\delta}(0)} \gamma(\mathbf{z}) d\mathbf{z} \left(\int_{\Omega} \int_{\Omega} \gamma u^2(\mathbf{x}) d\mathbf{x}' d\mathbf{x} - \int_{\Omega} \int_{\Omega} \gamma u(\mathbf{x}')u(\mathbf{x}) d\mathbf{x}' d\mathbf{x} \right), \end{aligned}$$

where

$$(3.3) \quad \int_{\Omega} \int_{\Omega} \gamma u^2(\mathbf{x}) d\mathbf{x}' d\mathbf{x} \leq \int_{B_{\delta}(0)} \gamma(\mathbf{z}) d\mathbf{z} \cdot \|u\|_{L^2(\Omega)}^2,$$

and

$$\begin{aligned} & \int_{\Omega} \int_{\Omega} \gamma u(\mathbf{x}')u(\mathbf{x}) d\mathbf{x}' d\mathbf{x} \\ &\leq \left(\int_{\Omega} \int_{\Omega} \gamma u^2(\mathbf{x}') d\mathbf{x}' d\mathbf{x} \right)^{1/2} \left(\int_{\Omega} \int_{\Omega} \gamma u^2(\mathbf{x}) d\mathbf{x}' d\mathbf{x} \right)^{1/2} \\ (3.4) \quad &\leq \int_{B_{\delta}(0)} \gamma(\mathbf{z}) d\mathbf{z} \cdot \|u\|_{L^2(\Omega)}^2, \end{aligned}$$

then the conclusion follows. $\square$

Lemma 3.2. *Assume the kernel function γ satisfies (2.9), then for any $u, v \in L_{n0}^2(\Omega)$,*

$$(3.5) \quad \int_{\Omega_s} v \mathcal{D}(\beta \mathcal{D}^*(u)) d\mathbf{x} - \int_{\Omega} \int_{\Omega} \mathcal{D}^*(v) \beta \mathcal{D}^*(u) d\mathbf{x}' d\mathbf{x} = 0.$$

Proof. For any $u, v \in L_{n0}^2(\Omega)$, considering Lemma 3.1 and nonlocal integration by part as shown in [21], we then have

$$\int_{\Omega} \int_{\Omega} \mathcal{D}^*(v) \beta \mathcal{D}^*(u) d\mathbf{x}' d\mathbf{x} = \int_{\Omega} v \mathcal{D}(\beta \mathcal{D}^*(u)) d\mathbf{x},$$

so (3.5) follows by considering constraint of v on Ω_c . $\square$

We may define a *nonlocal bilinear form* as follows

$$(3.6) \quad B(u, v) = \int_{\Omega} \int_{\Omega} \mathcal{D}^*(u) \beta \mathcal{D}^*(v) d\mathbf{x}' d\mathbf{x}.$$

for any $u, v \in V_c$.

Under the assumption (2.9), $\mathcal{D}(\beta \mathcal{D}^*(\cdot))$ defines a nonlocal diffusion operator from $L^2(\Omega)$ to $L^2(\Omega)$, and one expects to get the boundedness and coercivity of $B(\cdot, \cdot)$ in the space

$$(3.7) \quad V_c = L_{n0}^2(\Omega) := \{v \in L^2(\Omega_s) : v(\mathbf{x}) = 0 \text{ for } \mathbf{x} \in \Omega_c\},$$

that is, for all $u, v \in L_{n0}^2(\Omega)$, there exists a positive constant $C_1(\delta)$ such that

$$(3.8) \quad |B(u, v)| \leq 4M(\delta) \|u\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \quad \text{and} \quad B(u, u) \geq C_1(\delta) \|u\|_{L^2(\Omega)}^2.$$

We refer to [3, 28, 22, 5] for the detailed analysis on the validity of (3.7) and (3.8) for various kernel functions. For instance, [22] established (3.7) and (3.8) for a number of different combinations of domains and constraints and for a kernel function γ that is not only absolutely integrable but in fact, square integrable; in [28], the inequalities were shown for kernels that are absolutely integrable and uniformly bounded below by a positive constant everywhere; the case of non-negative integrable kernels with compact support was studied in detail in [5] while the inequalities were also shown in [3] for a number of cases covered by the space equivalence results discussed in [7].

With bilinear form (3.6), the problem (2.4) has the following weak formulation: find $u^* \in L_{n0}^2(\Omega)$, such that for any $v \in L_{n0}^2(\Omega)$,

$$(3.9) \quad B(u, v) = (f, v),$$

where $(f, v) = \int_{\Omega_s} f v d\mathbf{x}$.

For $u, v \in L_{n0}^2(\Omega)$, we aim to decompose the bilinear form

$$B(u, v) = \int_{\Omega} \int_{\Omega} \mathcal{D}^*(u) \beta \mathcal{D}^*(v) d\mathbf{x}' d\mathbf{x}$$

into the sum of local integrals over each element K to get a nonlocal a posteriori error estimator for the nonlocal diffusion equation (2.4).

3.2. Finite element discretization and a posteriori error estimate. To study the numerical solution of the linear nonlocal diffusion problem (2.4) by finite element methods, we let $\mathbf{S}^h$ be a finite dimensional subspace of $L_{n0}^2(\Omega)$, and $\mathbf{S}^h$ consists of piecewise smooth functions over $\{K\}$, where $\{K\}$ denotes the mesh of Ω_s ($\Omega_s = \cup K$).

By Lemma 3.5, we have for any $u, v \in L_{n0}^2(\Omega)$,

$$\begin{aligned} B(u, v) &= \int_{\Omega} \int_{\Omega} \mathcal{D}^*(u) \beta \mathcal{D}^*(v) d\mathbf{x}' d\mathbf{x} \\ &= \int_{\Omega} v(\mathbf{x}) \mathcal{D}(\beta \mathcal{D}^*(u)) d\mathbf{x} \end{aligned}$$

$$(3.10) \quad = \int_{\Omega_s} v(\mathbf{x}) \mathcal{D}(\beta \mathcal{D}^*(u)) d\mathbf{x}.$$

Let $u^h \in \mathbf{S}^h$ be the Galerkin approximation of u in $\mathbf{S}^h$ obtained using the weak form (3.9), then we have that

$$(3.11) \quad B(u^h, v^h) = (f, v^h), \quad \forall v^h \in \mathbf{S}^h.$$

By setting $e^h = u^* - u^h$ and using (3.10), we get for any $v \in L_{n_0}^2(\Omega)$,

$$(3.12) \quad \begin{aligned} B(e^h, v) &= B(u^*, v) - B(u^h, v) \\ &= \int_{\Omega_s} v(\mathbf{x}) (f - \mathcal{D}(\beta \mathcal{D}^*(u^h(\mathbf{x})))) d\mathbf{x} \\ &= (R^h, v), \end{aligned}$$

where

$$(3.13) \quad R^h(\mathbf{x}) = f(\mathbf{x}) - \mathcal{D}(\beta \mathcal{D}^*(u^h(\mathbf{x}))), \quad \forall \mathbf{x} \in K,$$

denotes the residual term.

Under assumption (2.9), for $v \in L_{n_0}^2(\Omega)$, define the *energy norm* as

$$(3.14) \quad \|v\|_E = \sqrt{B(v, v)},$$

since (3.8) implies the equivalence between $\|v\|_E$ and $\|v\|_{L^2(\Omega)}$, so we can define the *dual energy norm* as

$$(3.15) \quad \|v\|_{E^*} = \sup_{w \in L_{n_0}^2(\Omega)} \frac{(v, w)}{\|w\|_E},$$

and we have

Lemma 3.3. *For $u^*, u^h \in L_{n_0}^2(\Omega)$, the energy norm of error e^h equals the dual energy norm of the corresponding residual R^h , i.e.*

$$\|e^h\|_E = \|R^h\|_{E^*}.$$

Proof. First, we have by Cauchy-Schwartz inequality that for any $v \in L_{n_0}^2(\Omega)$,

$$(R^h, v) = B(e^h, v) \leq \|e^h\|_E \cdot \|v\|_E,$$

so

$$\|R^h\|_{E^*} = \sup_v \frac{(R^h, v)}{\|v\|_E} \leq \|e^h\|_E.$$

Second, since $e^h \in L_{n_0}^2(\Omega)$,

$$\|R^h\|_{E^*} = \sup_v \frac{(R^h, v)}{\|v\|_E} \geq \frac{B(e^h, e^h)}{\|e^h\|_E} = \|e^h\|_E,$$

then the conclusion follows. $\square$

We then have the following theorem for the a posteriori error approximation

Theorem 3.4. *Let*

$$(3.16) \quad \eta_\delta^h = \frac{\|R^h\|_{L^2(\Omega_s)}}{\sqrt{M(\delta)}}$$

denote the a posteriori error estimator. For fixed δ , it satisfies that

$$(3.17) \quad \frac{1}{2} \eta_\delta^h \leq \|e^h\|_E \leq \sqrt{\frac{M(\delta)}{C_1}} \eta_\delta^h.$$

Proof. By (3.8), we have for any $v \in L^2_{n0}(\Omega)$,

$$\sqrt{C_1(\delta)} \|v\|_{L^2(\Omega)} \leq \|v\|_E \leq 2\sqrt{M(\delta)} \|v\|_{L^2(\Omega)},$$

using the definition of $\|\cdot\|_{E^*}$ and Cauchy-Schwartz inequality, we obtain

$$\frac{1}{2\sqrt{M(\delta)}} \|v\|_{L^2(\Omega)} \leq \|v\|_{E^*} \leq \frac{1}{\sqrt{C_1(\delta)}} \|v\|_{L^2(\Omega)},$$

let $v = R^h$ in the above inequality, applying Lemma 3.3 gives the conclusion. $\square$

Remark 3.5. It is worthy noticing that in Theorem 3.4, we can provide $\|e^h\|_E$ both the upper and lower bounds, up to constant factors, in terms of η_δ^h . This is largely a consequence of the boundedness of the nonlocal diffusion operator and the solution operator in the energy space. It is interesting to examine if the constant factors appearing in the bounds are tight with respect to changes in the model parameters. The numerical experiments given later indicate that, for small δ , the lower bound is tight, while the upper bound fails to give a tight bound. This is not surprising as in the limit $\delta \rightarrow 0$, the nonlocal problems have been shown to reduce to their local counterparts [21, 23, 39] and for classical differential equations, a posteriori estimators take on different forms.

3.3. Relation with the local case. For simplicity, we assume Ω_c is a layered strip, with thickness δ , surrounding Ω_s , see Figure 1. Since the solution of the

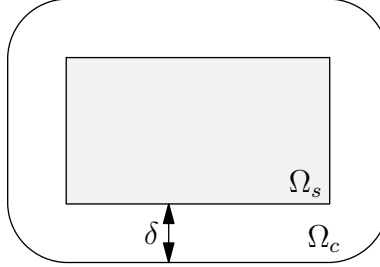


FIGURE 1. Ω_s and Ω_c .

problem (2.4) over Ω converges to the local limit, that is, the classic differential boundary value problem (2.10) over Ω_s as $\delta \rightarrow 0$ [21, 23, 39], with

$$(3.18) \quad \mathbb{D} = \lim_{\delta \rightarrow 0} \mathbb{D}_\delta,$$

where $(\mathbb{D}_\delta)_{ij} = \int_{B_\delta(0)} \gamma z_i z_j d\mathbf{z}$ for $i, j = 1, 2, \dots, d$, $\mathbf{z} = (z_1, z_2, \dots, z_d)^T \in \mathbb{R}^d$. It is thus natural to study if the a posteriori error estimator derived earlier for a nonlocal problem would converge to the corresponding estimator of its local limit in some suitable sense.

To study the limiting behavior as $\delta \rightarrow 0$ of the a posteriori error estimator on a given mesh, we assume that δ is sufficiently small such $\delta \ll h_K$, in the remaining discussion provided in this section. For the finite element analysis carried out here, we focus on the case of having globally continuous and piecewise smooth basis functions so that a conforming Galerkin approximation of the PDE model is given in the limit with the corresponding a posteriori error estimation briefly reviewed in

the previous section. More general analysis involving those based on discontinuous basis functions will be given in the future.

To balance the generality of analysis and the simplicity of notation, we further assume that, in this sub-section, u is defined over the tessellation $\{K\}$ such that

$$(3.19) \quad u \in C^0(\Omega) \cap H^1(\Omega), \quad u|_{\Omega_c} = 0, \quad u_K \in C^\tau(K) \text{ for any } K,$$

where $\tau \geq 2$, $u_\chi(\cdot)$ denotes the restriction of $u(\cdot)$ over any set χ . We impose the same assumption on v . Be aware that u does not need to solve weak form (3.9).

For any $K \subset \Omega$, Γ_K is defined by

$$(3.20) \quad \Gamma_K := \{\mathbf{x} \notin K \mid \text{Dist}(\mathbf{x}, K) \leq \delta\},$$

then it follows that

$$(3.21) \quad \begin{aligned} - \int_K v(\mathbf{x}) \mathcal{D}(\beta \mathcal{D}^*(u)) d\mathbf{x} &= \int_K v(\mathbf{x}) \int_K 2\gamma \cdot (u(\mathbf{x}') - u(\mathbf{x})) d\mathbf{x}' d\mathbf{x} \\ &+ \int_K v(\mathbf{x}) \int_{\Gamma_K} 2\gamma \cdot (u(\mathbf{x}') - u(\mathbf{x})) d\mathbf{x}' d\mathbf{x}. \end{aligned}$$

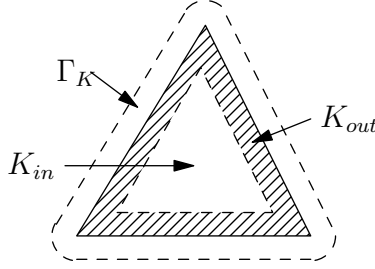


FIGURE 2. $K = K_{in} \cup K_{out}$, an illustration in the two-dimensional space.

Denote $K = K_{in} \cup K_{out}$, where $K_{out} = K \cap B_\delta(\partial K)$ (the δ -neighborhood of ∂K), while $K_{in} = K/K_{out}$ (see Figure 2 for an illustration of the two-dimensional case). When $\mathbf{x}', \mathbf{x} \in K$, since u is smooth within an element K , by Taylor expansion, we get

$$u(\mathbf{x}') = u(\mathbf{x}) + \nabla u(\mathbf{x}) \cdot (\mathbf{x}' - \mathbf{x}) + \frac{1}{2}(\mathbf{x}' - \mathbf{x})^T H_u(\mathbf{x})(\mathbf{x}' - \mathbf{x}) + o(\delta^2),$$

where H_u denotes the Hessian of u . Then the first term in (3.21) becomes

$$(3.22) \quad \begin{aligned} &\int_K v(\mathbf{x}) \int_K 2\gamma \cdot (u(\mathbf{x}') - u(\mathbf{x})) d\mathbf{x}' d\mathbf{x} \\ &= \int_K v(\mathbf{x}) \nabla u(\mathbf{x}) \cdot \int_K 2\gamma \cdot (\mathbf{x}' - \mathbf{x}) d\mathbf{x}' d\mathbf{x} \\ &\quad + \int_K v(\mathbf{x}) \int_K \gamma \cdot (\mathbf{x}' - \mathbf{x})^T H_u(\mathbf{x})(\mathbf{x}' - \mathbf{x}) d\mathbf{x}' d\mathbf{x} \\ &\quad + \int_K v(\mathbf{x}) \int_K 2\gamma \cdot o(\delta^2) d\mathbf{x}' d\mathbf{x} \\ &= T_1 + T_2 + T_3. \end{aligned}$$

Since u is piecewise smooth for $\mathbf{x} \in K$ and $\mathbf{x}' \in \Gamma_K$ respectively, we define $\mathbf{s}$ as the intersection of ∂K and the vector $(\mathbf{x}' - \mathbf{x})$ (as shown in Figure 4), one then has the following Taylor expansion

$$u_{\Gamma_K}(\mathbf{x}') = u(\mathbf{x}) + \nabla u_K(\mathbf{x}) \cdot (\mathbf{s} - \mathbf{x}) + \nabla u_{\Gamma_K}(\mathbf{s}) \cdot (\mathbf{x}' - \mathbf{s}) + o(\delta).$$

Then, the second term in (3.21) can be computed as following

$$\begin{aligned}
 & \int_K v(\mathbf{x}) \int_{\Gamma_K} 2\gamma \cdot (u(\mathbf{x}') - u(\mathbf{x})) d\mathbf{x}' d\mathbf{x} \\
 &= \int_{K_{out}} v(\mathbf{x}) \int_{\Gamma_K} 2\gamma \cdot (\nabla u_K(\mathbf{x}) \cdot (\mathbf{s} - \mathbf{x}) + \nabla u_{\Gamma_K}(\mathbf{s}) \cdot (\mathbf{x}' - \mathbf{s})) d\mathbf{x}' d\mathbf{x} \\
 & \quad + \int_{K_{out}} v(\mathbf{x}) \int_{\Gamma_K} 2\gamma \cdot o(\delta) d\mathbf{x}' d\mathbf{x} \\
 &= \int_{K_{out}} v(\mathbf{x}) \nabla u_K(\mathbf{x}) \cdot \int_{\Gamma_K} 2\gamma \cdot (\mathbf{x}' - \mathbf{x}) d\mathbf{x}' d\mathbf{x} \\
 & \quad + \int_{K_{out}} v(\mathbf{x}) \int_{\Gamma_K} \nabla u_K(\mathbf{x}) \cdot 2\gamma \cdot (\mathbf{s} - \mathbf{x}') d\mathbf{x}' d\mathbf{x} \\
 & \quad + \int_{K_{out}} v(\mathbf{x}) \int_{\Gamma_K} \nabla u_{\Gamma_K}(\mathbf{s}) \cdot 2\gamma \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\mathbf{x} \\
 & \quad + \int_{K_{out}} v(\mathbf{x}) \int_{\Gamma_K} 2\gamma \cdot o(\delta) d\mathbf{x}' d\mathbf{x} \\
 &= \int_{K_{out}} v(\mathbf{x}) \nabla u_K(\mathbf{x}) \cdot \int_{\Gamma_K} 2\gamma \cdot (\mathbf{x}' - \mathbf{x}) d\mathbf{x}' d\mathbf{x} \\
 & \quad + \int_{K_{out}} v(\mathbf{x}) \int_{\Gamma_K} (\nabla u_{\Gamma_K}(\mathbf{s}) - \nabla u_K(\mathbf{x})) \cdot 2\gamma \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\mathbf{x} \\
 & \quad + \int_{K_{out}} v(\mathbf{x}) \int_{\Gamma_K} 2\gamma \cdot o(\delta) d\mathbf{x}' d\mathbf{x} \\
 (3.23) \quad &= S_1 + S_2 + S_3.
 \end{aligned}$$

Next we will try to evaluate $\{T_i, S_i\}$, $i = 1, 2, 3$. First, by the symmetry of γ , we get

$$\begin{aligned}
 T_1 + S_1 &= \int_K v(\mathbf{x}) \nabla u_K(\mathbf{x}) \cdot \int_{K \cup \Gamma_K} 2\gamma \cdot (\mathbf{x}' - \mathbf{x}) d\mathbf{x}' d\mathbf{x} \\
 &= \int_K v(\mathbf{x}) \nabla u_K(\mathbf{x}) \cdot \int_{B_\delta(\mathbf{x})} 2\gamma \cdot (\mathbf{x}' - \mathbf{x}) d\mathbf{x}' d\mathbf{x} \\
 (3.24) \quad &= 0.
 \end{aligned}$$

It is also easy to see that with assumption (2.9),

$$(3.25) \quad T_3 \rightarrow 0, \quad S_3 \rightarrow 0 \quad \text{as} \quad \delta \rightarrow 0.$$

It now only remains to evaluate T_2 and S_2 . Define

$$\widehat{T}_2 = \int_{K_{out}} v(\mathbf{x}) \int_K \gamma \cdot (\mathbf{x}' - \mathbf{x})^T H_u(\mathbf{x})(\mathbf{x}' - \mathbf{x}) d\mathbf{x}' d\mathbf{x},$$

and it follows that

$$\widehat{T}_2 \rightarrow 0 \quad \text{as} \quad \delta \rightarrow 0.$$

Let us substitute $(\mathbf{x}' - \mathbf{x})$ by $\mathbf{z} = (z_1, z_2, \dots, z_d)^T$, then

$$\begin{aligned}
 T_2 &= \int_{K_{in}} v(\mathbf{x}) \int_K \gamma \cdot (\mathbf{x}' - \mathbf{x})^T H_u(\mathbf{x}) (\mathbf{x}' - \mathbf{x}) d\mathbf{x}' d\mathbf{x} + \widehat{T}_2 \\
 &= \int_{K_{in}} v(\mathbf{x}) \int_{B_\delta(0)} \gamma \cdot \mathbf{z}^T H_u(\mathbf{x}) \mathbf{z} d\mathbf{z} d\mathbf{x} + \widehat{T}_2 \\
 &= \int_{K_{in}} v(\mathbf{x}) \int_{B_\delta(0)} \gamma \cdot \left(\sum_{i,j} z_i z_j u_{ij}(\mathbf{x}) \right) d\mathbf{z} d\mathbf{x} + \widehat{T}_2 \\
 (3.26) \quad &= \int_{K_{in}} \nabla \cdot (\mathbb{D}_\delta \nabla u(\mathbf{x})) v(\mathbf{x}) d\mathbf{x} + \widehat{T}_2,
 \end{aligned}$$

where u_{ij} denotes the partial derivatives of u along x_i and x_j directions respectively, $\mathbb{D}_\delta$ is an $d \times d$ symmetric matrix defined by

$$(\mathbb{D}_\delta)_{ij} = \int_{B_\delta(0)} \gamma z_i z_j d\mathbf{z}.$$

Finally,

$$(3.27) \quad S_2 = \int_{K_{out}} v(\mathbf{x}) \int_{\Gamma_K} 2\gamma (\nabla u_{\Gamma_K}(\mathbf{s}) - \nabla u_K(\mathbf{x})) \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\mathbf{x}.$$

For the convenience of later computations, since δ is sufficiently small we rewrite the above equation as

$$\begin{aligned}
 S_2 &= \sum_{\mathbf{r} \in \text{edge}(K)} \int_{K_{out} \cap B_\delta(\mathbf{r})} v(\mathbf{x}) \int_{\Gamma_K \cap B_\delta(\mathbf{r})} 2\gamma \cdot (\nabla u_{\Gamma_K}(\mathbf{s}) \\
 (3.28) \quad &\quad - \nabla u_K(\mathbf{x})) \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\mathbf{x} + o(1),
 \end{aligned}$$

where $\text{edge}(K)$ denotes the set of boundary edges (faces) of K and $B_\delta(\mathbf{r})$ the δ -neighborhood of edge (face) $\mathbf{r}$.

Suppose that elements K and K' have a common edge (face) $\mathbf{e}$. If we repeat the above computation on K' , we can get the terms $T'_1, T'_2, T'_3, S'_1, S'_2$ and S'_3 corresponding to T_1, T_2, T_3, S_1, S_2 and S_3 , respectively, among which

$$\begin{aligned}
 S'_2 &= \sum_{\mathbf{r} \in \text{edge}(K')} \int_{K'_{out} \cap B_\delta(\mathbf{r})} v(\mathbf{x}) \int_{\Gamma_{K'} \cap B_\delta(\mathbf{r})} 2\gamma \cdot (\nabla u_{\Gamma_{K'}}(\mathbf{s}) \\
 (3.29) \quad &\quad - \nabla u_{K'}(\mathbf{x})) \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\mathbf{x} + o(1).
 \end{aligned}$$

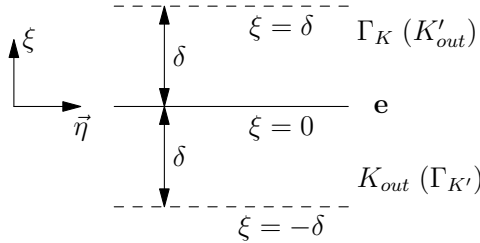


FIGURE 3. The δ -neighborhood of an edge $\mathbf{e}$.

Without loss of generality, assume the outward normal of K on $\mathbf{e}$ is $\vec{\mathbf{n}}_{\mathbf{e}} = (1, 0, 0, \dots, 0)^T \in \mathbb{R}^d$. Here we use the coordinate system $\xi \times \vec{\eta}$, where $\xi \in \mathbb{R}$ and

$\vec{\eta} \in \mathbb{R}^{d-1}$, as shown in Figure 3, i.e., edge (or face) $\mathbf{e}$ is on the $(d-1)$ -dimensional plane $\xi = 0$.

One can easily observe that

$$(3.30) \quad (\mathbf{e} \times [-\delta, 0]) \approx K_{out} \cap B_\delta(\mathbf{e}) \approx \Gamma_{K'} \cap B_\delta(\mathbf{e}),$$

the difference between any two of the above sets is of magnitude $O(\delta^d)$, which means the difference can be ignored in limit sense. Similarly, we can also have

$$(3.31) \quad (\mathbf{e} \times [0, \delta]) \approx K'_{out} \cap B_\delta(\mathbf{e}) \approx \Gamma_K \cap B_\delta(\mathbf{e}),$$

where the difference between any two sets again can be ignored in limit sense.

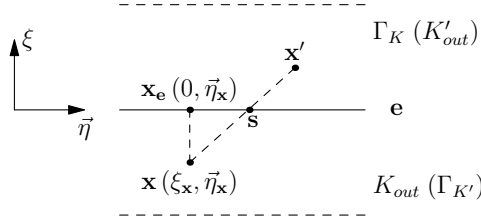


FIGURE 4. A projection of $\mathbf{x}$ onto $\mathbf{e}$.

Then let us add (3.28) and (3.29), and evaluate the contribution from edge(or face) $\mathbf{e}$ as following (drop the high order terms):

$$\begin{aligned}
 & \int_{K_{out} \cap B_\delta(\mathbf{e})} v(\mathbf{x}) \int_{\Gamma_K \cap B_\delta(\mathbf{e})} 2\gamma \cdot (\nabla u_{\Gamma_K}(\mathbf{s}) - \nabla u_K(\mathbf{x})) \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\mathbf{x} \\
 & + \int_{K'_{out} \cap B_\delta(\mathbf{e})} v(\mathbf{x}) \int_{\Gamma_{K'} \cap B_\delta(\mathbf{e})} 2\gamma \cdot (\nabla u_{\Gamma_{K'}}(\mathbf{s}) - \nabla u_{K'}(\mathbf{x})) \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\mathbf{x} \\
 = & \int_{K_{out} \cap B_\delta(\mathbf{e})} v(\mathbf{x}) \int_{\Gamma_K \cap B_\delta(\mathbf{e})} 2\gamma \cdot (\nabla u_{\Gamma_K}(\mathbf{s}) - \nabla u_K(\mathbf{x})) \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\mathbf{x} \\
 & + \int_{\Gamma_{K'} \cap B_\delta(\mathbf{e})} \int_{K'_{out} \cap B_\delta(\mathbf{e})} 2\gamma \cdot v(\mathbf{x}') (\nabla u_{\Gamma_{K'}}(\mathbf{s}) - \nabla u_{K'}(\mathbf{x}')) \cdot (\mathbf{x} - \mathbf{s}) d\mathbf{x}' d\mathbf{x} \\
 = & \int_{\mathbf{e}} \int_{-\delta}^0 \int_{B_\delta(\mathbf{x}) \cap \Gamma_K} 2\gamma \cdot v(\mathbf{x}) (\nabla u_{\Gamma_K}(\mathbf{s}) - \nabla u_K(\mathbf{x})) \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\xi d\vec{\eta} \\
 & + \int_{\mathbf{e}} \int_{-\delta}^0 \int_{B_\delta(\mathbf{x}) \cap K'_{out}} 2\gamma \cdot v(\mathbf{x}') (\nabla u_{K'}(\mathbf{x}') - \nabla u_{\Gamma_{K'}}(\mathbf{s})) \cdot (\mathbf{s} - \mathbf{x}) d\mathbf{x}' d\xi d\vec{\eta},
 \end{aligned}
 \tag{3.32}$$

where the second to the last step is obtained by switching order of integration; the last step is obtained by using (3.30) and the fact that $\mathbf{x}'$ interacts with $\mathbf{x}$ only if $\mathbf{x}' \in B_\delta(\mathbf{x})$. Note that in the above derivation we let $\mathbf{x} = (\xi_{\mathbf{x}}, \vec{\eta}_{\mathbf{x}})$, substitute K'_{out} by Γ_K , $u_{K'}$ by u_{Γ_K} , $u_{\Gamma_{K'}}$ by u_K in (3.32), and denote by $\mathbf{x}_{\mathbf{e}} = (0, \vec{\eta}_{\mathbf{x}})$ the projection of $\mathbf{x}$ on $\mathbf{e}$, as shown in Figure 4.

Since $\mathbf{x}$, $\mathbf{x}'$ and $\mathbf{s}$ are all in the δ -neighborhood of $\mathbf{x}_{\mathbf{e}}$, we can perform the Taylor expansion for $v(\mathbf{x})$, $v(\mathbf{x}')$, $\nabla u_{\Gamma_K}(\mathbf{s})$, $\nabla u_{\Gamma_K}(\mathbf{x}')$, $\nabla u_K(\mathbf{x})$ and $\nabla u_K(\mathbf{s})$ at point $\mathbf{x}_{\mathbf{e}}$ respectively, it follows that (3.32) can be approximated as, after dropping the high

order terms,

$$\begin{aligned}
 & \int_{\mathbf{e}} v(\mathbf{x}_{\mathbf{e}}) (\nabla u_{\Gamma_K}(\mathbf{x}_{\mathbf{e}}) - \nabla u_K(\mathbf{x}_{\mathbf{e}})) \int_{-\delta}^0 \int_{B_{\delta}(\mathbf{x}) \cap \Gamma_K} 2\gamma \cdot (\mathbf{x}' - \mathbf{x}) d\mathbf{x}' d\xi d\vec{\eta} \\
 &= \int_{\mathbf{e}} v(\mathbf{x}_{\mathbf{e}}) (\nabla u_{\Gamma_K}(\mathbf{x}_{\mathbf{e}}) - \nabla u_K(\mathbf{x}_{\mathbf{e}})) \int_{-\delta}^0 \int_{SC(\delta, \delta+\xi)} 2\gamma \mathbf{z} d\mathbf{z} d\xi d\vec{\eta},
 \end{aligned}
 \tag{3.33}$$

where $\mathbf{z} = \mathbf{x}' - \mathbf{x}$, the integral domain $SC(\delta, \delta + \xi)$ denotes the spherical cap of a sphere with radius δ and height $\delta + \xi$, as shown in Figure 5.

From (3.33) we obtain the jump of ∇u on interior edge $\mathbf{e}$, and it remains to show that the integral

$$\int_{-\delta}^0 \int_{SC(\delta, \delta+\xi)} 2\gamma \mathbf{z} d\mathbf{z} d\xi
 \tag{3.34}$$

actually converges to outward normal with the same coefficient as the Laplacian term in (3.26).

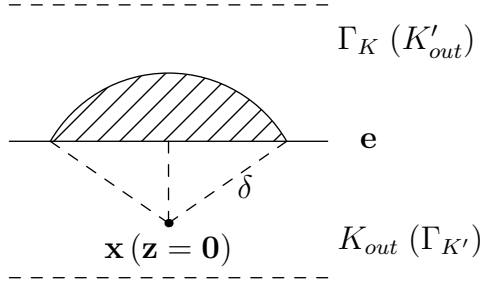


FIGURE 5. An illustration of the integral domain $SC(\delta, \delta + \xi)$ that denotes the spherical cap of a sphere with radius δ and height $\delta + \xi$.

Let $\mathbf{z} = (z_1, z_2, \dots, z_d)^T$, we switch the order of the integration, where we omit some details, in (3.34) and obtain the following result for fixed $\vec{\eta}_{\mathbf{x}}$,

$$\begin{aligned}
 \int_{-\delta}^0 \int_{SC(\delta, \delta+\xi)} 2\gamma \mathbf{z} d\mathbf{z} d\xi &= \int_{SC(\delta, \delta)} \int_{-z_1}^0 2\gamma \mathbf{z} d\xi d\mathbf{z} \\
 &= \int_{SC(\delta, \delta)} 2\gamma z_1 \mathbf{z} d\mathbf{z} \\
 &= \int_{SC(\delta, \delta)} 2\gamma \cdot (z_1^2, z_1 z_2, z_1 z_3, \dots, z_1 z_d)^T d\mathbf{z} \\
 &= \int_{B_{\delta}(0)} \gamma \cdot (z_1^2, z_1 z_2, z_1 z_3, \dots, z_1 z_d)^T d\mathbf{z} \\
 &= \mathbb{D}_{\delta} \cdot \vec{\mathbf{n}}_{\mathbf{e}},
 \end{aligned}
 \tag{3.35}$$

where the second to last equality holds since the integration domain $SC(\delta, \delta)$ is actually a half-sphere.

When $\mathbf{e}$ is not horizontal, we can simply perform a rotation operator on $\mathbf{x}'$ and $\mathbf{x}$, then the similar result as in (3.35) follows.

Now we go back to (3.32), it is easy to see

$$\begin{aligned}
 & \int_{K_{out} \cap B_\delta(\mathbf{e})} v(\mathbf{x}) \int_{\Gamma_K \cap B_\delta(\mathbf{e})} 2\gamma \cdot (\nabla u_{\Gamma_K}(\mathbf{s}) - \nabla u_K(\mathbf{x})) \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\mathbf{x} + o(\delta) \\
 &= \int_{K'_{out} \cap B_\delta(\mathbf{e})} v(\mathbf{x}) \int_{\Gamma_{K'} \cap B_\delta(\mathbf{e})} 2\gamma \cdot (\nabla u_{\Gamma_{K'}}(\mathbf{s}) - \nabla u_{K'}(\mathbf{x})) \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\mathbf{x},
 \end{aligned}
 \tag{3.36}$$

and the limit of each integral exists as $\delta \rightarrow 0$.

Considering (3.28), (3.32), (3.33), (3.35) and (3.36), we have that as $\delta \rightarrow 0$,

$$(3.37) \quad S_2 \rightarrow -\frac{1}{2} \sum_{\mathbf{r} \in \text{edge}(K)} \int_{\mathbf{r}} v(\nu) [\mathbb{D}\nabla u(\nu)]_{\mathbf{r}} d\nu,$$

where $\mathbb{D} = \lim_{\delta \rightarrow 0} \mathbb{D}_\delta$, $[\mathbb{D}\nabla u(\cdot)]_{\mathbf{r}}$ denotes the jump of $\mathbb{D}\nabla u$ on $\mathbf{r}$.

Combining (3.21), (3.22), (3.23), (3.24), (3.25), (3.26) and (3.37), we obtain the following theorem

Theorem 3.6. *Under assumption (2.9), for u and v that satisfy assumption (3.19), as $\delta \rightarrow 0$ on any element K ,*

$$\begin{aligned}
 & - \int_K v(\mathbf{x}) \mathcal{D}(\beta \mathcal{D}^*(u)) d\mathbf{x} \\
 (3.38) \quad & \rightarrow \int_K v(\mathbf{x}) (\nabla \cdot (\mathbb{D}\nabla u(\mathbf{x}))) d\mathbf{x} - \frac{1}{2} \int_{\partial K} v(\nu) [\mathbb{D}\nabla u(\nu)]_{\mathbf{r}} d\nu.
 \end{aligned}$$

Consequently, we have

Corollary 3.7. *Under assumption (2.9), for u^h and v that satisfy assumption (3.19), as $\delta \rightarrow 0$ on any element K ,*

$$\begin{aligned}
 & \int_K v(\mathbf{x}) (f(\mathbf{x}) - \mathcal{D}(\beta \mathcal{D}^*(u^h))) d\mathbf{x} \\
 & \rightarrow \int_K v(\mathbf{x}) (f(\mathbf{x}) + \nabla \cdot (\mathbb{D}\nabla u^h(\mathbf{x}))) d\mathbf{x} - \frac{1}{2} \int_{\partial K} v(\nu) [\mathbb{D}\nabla u^h(\nu)]_{\mathbf{r}} d\nu,
 \end{aligned}$$

that is,

$$(3.39) \quad \int_K v(\mathbf{x}) R^h d\mathbf{x} \rightarrow \int_K v(\mathbf{x}) r_*^h d\mathbf{x} - \frac{1}{2} \int_{\partial K} v(\nu) R_*^h d\nu.$$

Remark 3.8. Notice that in Theorem 3.6, we actually proved the local limit of nonlocal operator on each element. Also, comparing (3.39) with (2.12), as $\delta \rightarrow 0$, the nonlocal residual-based a posteriori error approximation is consistent with the classical a posteriori error approximation (in the weak sense).

Summing up (3.38) for all the K 's, we would have as $\delta \rightarrow 0$,

$$\begin{aligned}
 & - \int_{\Omega} \int_{\Omega} \mathcal{D}^*(v) \beta \mathcal{D}^*(u) d\mathbf{x}' d\mathbf{x} = - \int_{\Omega_s} v(\mathbf{x}) \mathcal{D}(\beta \mathcal{D}^*(u)) d\mathbf{x} \\
 & \rightarrow \sum_K \left(\int_K v(\mathbf{x}) (\nabla \cdot (\mathbb{D}\nabla u(\mathbf{x}))) d\mathbf{x} - \frac{1}{2} \int_{\partial K} v(\nu) [\mathbb{D}\nabla u(\nu)]_{\mathbf{r}} d\nu \right) \\
 & = - \int_{\Omega_s} \nabla u(\mathbf{x}) \cdot \mathbb{D} \cdot \nabla v(\mathbf{x}) d\mathbf{x},
 \end{aligned}$$

which implies the nonlocal diffusion constraint value problem reduces to classical Poisson equation with Dirichlet boundary condition, as stated in [21, 28].

4. PERIDYNAMIC EQUATIONS

Under the assumption (2.9), we define the variational form of problem (2.8): find $\mathbf{u}^* \in \mathbf{L}_{n0}^2(\Omega) = (L_{n0}^2(\Omega))^d$, such that for any $\mathbf{v} \in \mathbf{L}_{n0}^2(\Omega)$,

$$(4.1) \quad \mathbf{B}(\mathbf{u}^*, \mathbf{v}) = (\mathbf{f}, \mathbf{v})$$

where the bilinear form is given by

$$\mathbf{B}(\mathbf{u}, \mathbf{v}) = \int_{\Omega} \int_{\Omega} \mathcal{G}^*(\mathbf{u}) \beta \mathcal{G}^*(\mathbf{v}) d\mathbf{x}' d\mathbf{x},$$

and

$$(\mathbf{f}, \mathbf{v}) = \int_{\Omega_s} \mathbf{f} \cdot \mathbf{v} d\mathbf{x}.$$

For the simplicity of analysis, we study the isotropic and homogeneous kernel function instead of the more general anisotropic kernels considered earlier in the case of nonlocal diffusion equations, so that we may avoid complicated notations. We note that there is no essential difference in the final conclusions between the isotropic and anisotropic cases. Thus, let us assume that β is isotropic and homogeneous, that is, a function of $|\mathbf{x}' - \mathbf{x}|$. Moreover, for a fixed $\delta > 0$,

$$\beta(\mathbf{x}', \mathbf{x}) = \begin{cases} \beta(|\mathbf{x}' - \mathbf{x}|) & \text{when } |\mathbf{x}' - \mathbf{x}| < \delta, \\ 0 & \text{otherwise.} \end{cases}$$

We set

$$\alpha(\mathbf{x}', \mathbf{x}) = \frac{\mathbf{x}' - \mathbf{x}}{|\mathbf{x}' - \mathbf{x}|},$$

and define the kernel function for PD model with $\mathbf{z} = \mathbf{x}' - \mathbf{x}$,

$$(4.2) \quad \Lambda(\mathbf{z}) = (\alpha \otimes \alpha \beta)(\mathbf{z}) = \frac{\mathbf{z} \otimes \mathbf{z}}{|\mathbf{z}|^2} \beta(|\mathbf{z}|),$$

which is also called the *micromodulus function* [24, 33].

With the above α and β , we can write (2.9) as

$$(4.3) \quad 0 < c(\delta) \leq \int_{B_\delta(0)} \beta(|\mathbf{z}|) d\mathbf{z} \leq M(\delta) < \infty.$$

In [23], the well-posedness of linear state-based and bond-based PD models with solutions in $\mathbf{L}^2(\Omega) = (L^2(\Omega))^d$ has been established for general symmetric square integrable kernel functions by showing the corresponding nonlocal bilinear form being continuous and coercive in the volume constrained space $(L_{n0}^2(\Omega))^d$. The coercivity is a particular consequence of spectral properties of the Hilbert-Schmidt operator and the precise characterization of the energy space. These results generalized similar properties proved under more special nonlocal boundary conditions in [39] with isotropic and homogeneous L^1 kernel functions. For the constrained volume condition considered here, similar to [3, 7], it is also possible to establish similar properties for the bilinear form with other kernels, say those homogeneous, isotropic and L^1 integrable kernels. In this paper, we focus on the finite element approximation by assuming the continuity and coercivity in the constrained energy space $(L_{n0}^2(\Omega))^d$ of the bilinear form corresponding to the linear bond-based PD

model. That is, for any $\mathbf{u}, \mathbf{v} \in \mathbf{L}_{n0}^2(\Omega)$, we assume that there exist non-negative constants C_2 and $C_3(\delta)$ such that

$$(4.4) \quad |\mathbf{B}(\mathbf{u}, \mathbf{v})| \leq C_2 M(\delta) \|\mathbf{u}\|_{\mathbf{L}^2(\Omega)} \|\mathbf{v}\|_{\mathbf{L}^2(\Omega)} \quad \text{and} \quad \mathbf{B}(\mathbf{u}, \mathbf{v}) \geq C_3(\delta) \|\mathbf{u}\|_{\mathbf{L}^2(\Omega)}^2.$$

Then, similar to the discussion earlier, we may state the following lemma about nonlocal green's identity,

Lemma 4.1. *Assume the kernel function Λ be defined as in (4.2), with α and β satisfying (2.9). Then for any $\mathbf{u}, \mathbf{v} \in \mathbf{L}_{n0}^2(\Omega)$,*

$$(4.5) \quad \int_{\Omega_s} \mathbf{v} \mathcal{G}(\beta \mathcal{G}^*(\mathbf{u})) d\mathbf{x} - \int_{\Omega} \int_{\Omega} \mathcal{G}^*(\mathbf{v}) \beta \mathcal{G}^*(\mathbf{u}) d\mathbf{x}' d\mathbf{x} = 0.$$

We note that such results have been established in [21] with the implicit assumption that the integrands are well defined. By the properties of the bilinear form and thus the PD operator, the identity can be stated in the suitable spaces. As $\delta \rightarrow 0$, the problem (2.8) over Ω converges to *Navier-Lamé* equation over Ω_s with homogeneous Dirichlet constrained volume condition [23, 39]:

$$(4.6) \quad \begin{cases} -\mu \nabla \cdot (\nabla \mathbf{u}(\mathbf{x}) + \nabla^T \mathbf{u}(\mathbf{x})) - \mu \nabla (\nabla \cdot \mathbf{u}(\mathbf{x})) = \mathbf{f}(\mathbf{x}) & \text{for } \mathbf{x} \in \Omega_s \\ \mathbf{u}(\mathbf{x}) = 0 & \text{for } \mathbf{x} \in \partial\Omega_s, \end{cases}$$

with the coefficient μ , under the above assumption for Λ , can be evaluated as

$$(4.7) \quad \mu = \lim_{\delta \rightarrow 0} \int_{B_\delta(0)} \frac{\beta}{|\mathbf{z}|^2} z_1^2 z_2^2 d\mathbf{z},$$

where $\mathbf{z} = (z_1, \dots, z_d)^T$.

The weak form of problem (4.6) can be formulated as: find $\mathbf{u} \in \mathbf{H}_0^1(\Omega) (= \mathbf{L}_0^2(\Omega) \cap (H^1(\Omega))^d)$ such that for any $\mathbf{v} \in \mathbf{H}_0^1(\Omega)$

$$(4.8) \quad \mu(\nabla \mathbf{u} + \nabla^T \mathbf{u} : \nabla \mathbf{v}) + \mu(\nabla \cdot \mathbf{u}, \nabla \cdot \mathbf{v}) = (\mathbf{f}, \mathbf{v}),$$

where the symbol $\cdot$ denotes the *Frobenius product*.

4.1. A posteriori error analysis. Using a similar finite element discretization as in the last section, we denote $\mathbf{u}^h \in (\mathbf{S}_{local}^h)^d$ the finite element approximation of $\mathbf{u}$ by solving (4.8). Applying the framework in [37], we can get a residual-type identity that states, for any $\mathbf{v} \in \mathbf{H}_0^1(\Omega)$,

$$(4.9) \quad \begin{aligned} & \mu((\nabla + \nabla^T)(\mathbf{u} - \mathbf{u}^h) : \nabla \mathbf{v}) + \mu(\nabla \cdot (\mathbf{u} - \mathbf{u}^h), \nabla \cdot \mathbf{v}) \\ &= \sum_K \left(\int_K (\mathbf{f} + \mu \nabla \cdot (\nabla \mathbf{u}^h + \nabla^T \mathbf{u}^h) + \mu \nabla (\nabla \cdot \mathbf{u}^h)) \cdot \mathbf{v} d\mathbf{x} \right. \\ & \quad \left. - \frac{1}{2} \int_{\partial K} \mu [\nabla \mathbf{u}^h + \nabla^T \mathbf{u}^h + (\nabla \cdot \mathbf{u}^h) \mathbf{I}]_{\mathbf{r}} \cdot \mathbf{v} d\nu \right), \end{aligned}$$

where $[\nabla \mathbf{u}^h + \nabla^T \mathbf{u}^h + (\nabla \cdot \mathbf{u}^h) \mathbf{I}]_{\mathbf{r}}$ denotes the normal flux on an edge (face) $\mathbf{r}$ defined as follows: suppose $\mathbf{r}$ is the common edge(face) of elements K and K' , then

$$(4.10) \quad \begin{aligned} [\nabla \mathbf{u}^h + \nabla^T \mathbf{u}^h + (\nabla \cdot \mathbf{u}^h) \mathbf{I}]_{\mathbf{r}} &= (\nabla \mathbf{u}_K^h + \nabla^T \mathbf{u}_K^h + (\nabla \cdot \mathbf{u}_K^h) \mathbf{I}) \cdot \vec{\mathbf{n}}_K \\ & \quad + (\nabla \mathbf{u}_{K'}^h + \nabla^T \mathbf{u}_{K'}^h + (\nabla \cdot \mathbf{u}_{K'}^h) \mathbf{I}) \cdot \vec{\mathbf{n}}_{K'}. \end{aligned}$$

We now can define the *interior residual*, similarly as for the scalar case discussed earlier,

$$(4.11) \quad \mathbf{r}_*^h = \mathbf{f} + \mu \nabla \cdot (\nabla \mathbf{u}^h + \nabla^T \mathbf{u}^h) + \mu \nabla (\nabla \cdot \mathbf{u}^h) \quad \text{in } K,$$

and the *boundary residual*

$$(4.12) \quad \mathbf{R}_*^h = \mu \llbracket \nabla \mathbf{u}^h + \nabla^T \mathbf{u}^h + (\nabla \cdot \mathbf{u}^h) \mathbf{I} \rrbracket_{\mathbf{r}} \quad \text{on any interior edge (face) } \mathbf{r},$$

and then obtain the local a posteriori error estimator correspondingly.

Following the steps in the last section, we get the residual for the peridynmaic problem (2.8)

$$(4.13) \quad \mathbf{R}^h(\mathbf{x}) = \mathbf{f}(\mathbf{x}) - \mathcal{G}(\beta \mathcal{G}^*(\mathbf{u}^h(\mathbf{x}))), \quad \forall \mathbf{x} \in K,$$

where $\mathbf{u}^h \in (\mathbf{S}^h)^d$ is the finite element approximation of $\mathbf{u}$ by solving (4.1). The reliability and efficiency also can be proved as in the previous section. We therefore are able to get the following theorem

Theorem 4.2. *Suppose that $\mathbf{u}^* \in \mathbf{L}_{n0}^2(\Omega)$ is the weak solution of (4.1), $\mathbf{u}^h \in (\mathbf{S}^h)^d$ is the finite element approximation of $\mathbf{u}$, $\mathbf{R}^h$ is defined as in (4.13), $\mathbf{e}^h = \mathbf{u}^* - \mathbf{u}^h$ is the exact error. Then there exists non-negative constants C_4 and $C_5(\delta)$ such that*

$$(4.14) \quad C_4 \boldsymbol{\eta}_\delta^h \leq \|\mathbf{e}^h\|_{\mathbf{E}} \leq C_5(\delta) \boldsymbol{\eta}_\delta^h,$$

where $\boldsymbol{\eta}_\delta^h = \|\mathbf{R}^h\|_{\mathbf{L}^2(\Omega)} / \sqrt{M(\delta)}$ denotes the a posteriori error estimator, $\|\cdot\|_{\mathbf{E}} = \sqrt{\mathbf{B}(\cdot, \cdot)}$ denotes the energy norm.

Next, we will mimic the analysis performed for nonlocal diffusion problem, and find the connection between (4.13) and (4.9).

4.2. Relation with local case. In this subsection, we use the assumption on Ω_c as in Section 3.3. And further assume

$$(4.15) \quad \begin{aligned} \mathbf{u}, \mathbf{v} &\in (C^0(\Omega) \cap H^1(\Omega))^d, \quad \mathbf{u}|_{\Omega_c} = 0, \quad \mathbf{v}|_{\Omega_c} = 0, \\ \mathbf{u}_K, \mathbf{v}_K &\in (C^\tau(K))^d \text{ for any } K, \text{ where } \tau \geq 2. \end{aligned}$$

Consider the bilinear form in (4.1). Applying Lemma 4.1 gives

$$(4.16) \quad \begin{aligned} & - \int_{\Omega} \int_{\Omega} \mathcal{G}^*(\mathbf{u}) \beta \mathcal{G}^*(\mathbf{v}) d\mathbf{x}' d\mathbf{x} \\ & = - \int_{\Omega_s} \mathbf{v}(\mathbf{x}) \cdot \mathcal{G}(\beta \mathcal{G}^*(\mathbf{u})) d\mathbf{x}' d\mathbf{x} \\ & = - \sum_K \int_K \mathbf{v}(\mathbf{x}) \cdot \mathcal{G}(\beta \mathcal{G}^*(\mathbf{u})) d\mathbf{x}' d\mathbf{x} \\ & = \sum_K \int_K \mathbf{v}(\mathbf{x}) \cdot \int_K 2\Lambda \cdot (\mathbf{u}(\mathbf{x}') - \mathbf{u}(\mathbf{x})) d\mathbf{x}' d\mathbf{x} \\ & \quad + \sum_K \int_K \mathbf{v}(\mathbf{x}) \cdot \int_{\Gamma_K} 2\Lambda \cdot (\mathbf{u}(\mathbf{x}') - \mathbf{u}(\mathbf{x})) d\mathbf{x}' d\mathbf{x}. \end{aligned}$$

For any K , consider the first term in (4.16),

$$\begin{aligned} & \int_K \mathbf{v}(\mathbf{x}) \cdot \int_K 2\Lambda \cdot (\mathbf{u}(\mathbf{x}') - \mathbf{u}(\mathbf{x})) d\mathbf{x}' d\mathbf{x} \\ & = \int_K \mathbf{v}(\mathbf{x}) \cdot \int_K 2\Lambda \cdot \nabla \mathbf{u}(\mathbf{x}) \cdot (\mathbf{x}' - \mathbf{x}) d\mathbf{x}' d\mathbf{x} \\ & \quad + \int_K \mathbf{v}(\mathbf{x}) \cdot \int_K 2\Lambda \cdot \frac{1}{2} (\mathbf{x}' - \mathbf{x})^T H_{\mathbf{u}}(\mathbf{x}) \cdot (\mathbf{x}' - \mathbf{x}) d\mathbf{x}' d\mathbf{x} \end{aligned}$$

$$\begin{aligned}
& + \int_K \mathbf{v}(\mathbf{x}) \cdot \int_K 2\Lambda \cdot o(\delta^2) d\mathbf{x}' d\mathbf{x} \\
(4.17) \quad & = T_1 + T_2 + T_3,
\end{aligned}$$

where $H_{\mathbf{u}}(\cdot)$ is the Hessian tensor of $\mathbf{u}$.

Assume $K = K_{in} \cup K_{out}$, where K_{in} and K_{out} are defined as before. Denote $\mathbf{z} = \mathbf{x}' - \mathbf{x}$, and evaluate T_2 as

$$\begin{aligned}
& \int_K \mathbf{v}(\mathbf{x}) \cdot \int_K 2\Lambda \cdot \frac{1}{2} (\mathbf{x}' - \mathbf{x})^T H_{\mathbf{u}}(\mathbf{x}) (\mathbf{x}' - \mathbf{x}) d\mathbf{x}' d\mathbf{x} \\
(4.18) \quad & = \int_{K_{in}} \mathbf{v}(\mathbf{x}) \cdot \int_{B_\delta(0)} \frac{\beta}{|\mathbf{z}|^2} \mathbf{z} \otimes \mathbf{z} \cdot \mathbf{z}^T H_{\mathbf{u}}(\mathbf{x}) \mathbf{z} d\mathbf{z} d\mathbf{x} + o(1),
\end{aligned}$$

Let $\mathbf{z} = (z_1, z_2, \dots, z_d)^T$, $\mathbf{u} = (u_1, u_2, \dots, u_d)^T$, then we can compute the i -th component of $(\mathbf{z} \otimes \mathbf{z} \cdot \mathbf{z}^T H_{\mathbf{u}}(\mathbf{x}) \mathbf{z})$ as

$$(4.19) \quad (\mathbf{z} \otimes \mathbf{z} \cdot \mathbf{z}^T H_{\mathbf{u}}(\mathbf{x}) \mathbf{z})_i = \sum_{j,k,l} z_i z_j z_k z_l u_{l,jk}(\mathbf{x}),$$

where $u_{l,jk}$ denotes the second-order mixed derivative of u_l along x_j and x_k directions.

Using the symmetry of integral over $B_\delta(0)$, we have that the only non-zero terms from (4.19) after integration, are

$$\begin{aligned}
& \int_{B_\delta(0)} \frac{\beta}{|\mathbf{z}|^2} \left(z_i^4 u_{i,ii}(\mathbf{x}) + \sum_{j \neq i} z_i^2 z_j^2 (u_{i,jj}(\mathbf{x}) + u_{j,ij}(\mathbf{x}) + u_{j,ji}(\mathbf{x})) \right) d\mathbf{z} \\
& = \left(\int_{B_\delta(0)} \frac{\beta}{|\mathbf{z}|^2} z_1^2 z_2^2 d\mathbf{z} \right) (3u_{i,ii}(\mathbf{x}) + \sum_{j \neq i} (u_{i,jj}(\mathbf{x}) + u_{j,ij}(\mathbf{x}) + u_{j,ji}(\mathbf{x}))) \\
& = \left(\int_{B_\delta(0)} \frac{\beta}{|\mathbf{z}|^2} z_1^2 z_2^2 d\mathbf{z} \right) \sum_j (u_{i,jj}(\mathbf{x}) + u_{j,ij}(\mathbf{x}) + u_{j,ji}(\mathbf{x})) \\
(4.20) \quad & = \left(\int_{B_\delta(0)} \frac{\beta}{|\mathbf{z}|^2} z_1^2 z_2^2 d\mathbf{z} \right) (\nabla \cdot (\nabla \mathbf{u} + \nabla^T \mathbf{u}) + \nabla(\nabla \cdot \mathbf{u}))_i,
\end{aligned}$$

since we have the relationship

$$\int_{B_\delta(0)} \beta \frac{z_l^4}{|\mathbf{z}|^2} d\mathbf{z} = \int_{B_\delta(0)} \beta \frac{z_s^4}{|\mathbf{z}|^2} d\mathbf{z} = 3 \int_{B_\delta(0)} \beta \frac{z_m^2 z_n^2}{|\mathbf{z}|^2} d\mathbf{z} = 3 \int_{B_\delta(0)} \beta \frac{z_t^2 z_p^2}{|\mathbf{z}|^2} d\mathbf{z},$$

for any l, m, n ($m \neq n$), s, t, p ($t \neq p$) = 1, 2, ..., d .

Therefore, as $\delta \rightarrow 0$,

$$(4.21) \quad T_2 \rightarrow \int_{K_{in}} \mathbf{v}(\mathbf{x}) \cdot \mu (\nabla \cdot (\nabla \mathbf{u} + \nabla^T \mathbf{u}) + \nabla(\nabla \cdot \mathbf{u})) d\mathbf{x}.$$

For the second term in (4.16), we have

$$\begin{aligned}
& \int_K \mathbf{v}(\mathbf{x}) \cdot \int_{\Gamma_K} 2\Lambda \cdot (\mathbf{u}(\mathbf{x}') - \mathbf{u}(\mathbf{x})) d\mathbf{x}' d\mathbf{x} \\
& = \int_K \mathbf{v}(\mathbf{x}) \cdot \int_{\Gamma_K} 2\Lambda \cdot \nabla \mathbf{u}_K(\mathbf{x}) \cdot (\mathbf{x}' - \mathbf{x}) d\mathbf{x}' d\mathbf{x}
\end{aligned}$$

$$\begin{aligned}
& + \int_K \mathbf{v}(\mathbf{x}) \cdot \int_{\Gamma_K} 2\Lambda \cdot [\nabla \mathbf{u}_{\Gamma_K}(\mathbf{s}) - \nabla \mathbf{u}_K(\mathbf{x})] \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\mathbf{x} \\
& + \int_K \mathbf{v}(\mathbf{x}) \cdot \int_{\Gamma_K} 2\Lambda \cdot o(\delta) d\mathbf{x}' d\mathbf{x} \\
(4.22) \quad & = S_1 + S_2 + S_3.
\end{aligned}$$

Adding up T_1 and S_1 gives us

$$\begin{aligned}
T_1 + S_1 &= \int_K \mathbf{v}(\mathbf{x}) \cdot \int_{K \cup \Gamma_K} 2\Lambda \cdot \nabla \mathbf{u}_K(\mathbf{x}) \cdot (\mathbf{x}' - \mathbf{x}) d\mathbf{x}' d\mathbf{x} \\
&= \int_K \mathbf{v}(\mathbf{x}) \cdot \int_{B_\delta(0)} 2\Lambda \cdot \nabla \mathbf{u}(\mathbf{x}) \cdot \mathbf{z} d\mathbf{z} d\mathbf{x}, \\
(4.23) \quad &= 0.
\end{aligned}$$

By the symmetry of $B_\delta(0)$, it is also not hard to show (considering the scaling factor in β) that

$$(4.24) \quad T_3 + S_3 \rightarrow 0 \quad \text{as} \quad \delta \rightarrow 0.$$

Next, we will evaluate the term S_2 by partly mimicking the argument made in previous sections. Assume that the elements K and K' have a common edge $\mathbf{e}$ and the outward normal of K on $\mathbf{e}$ is $\tilde{\mathbf{n}}_{\mathbf{e}} = (1, 0, 0, \dots, 0)^T$. Since $\mathbf{v}(\cdot)$ is continuous across elements, we then have

$$\begin{aligned}
S_2 &= \int_K \mathbf{v}(\mathbf{x}) \cdot \int_{\Gamma_K} 2\Lambda \cdot (\nabla \mathbf{u}_{\Gamma_K}(\mathbf{s}) - \nabla \mathbf{u}_K(\mathbf{x})) \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\mathbf{x} \\
&= \sum_{\mathbf{r} \in \text{edge}(K)/\{\mathbf{e}\}} \int_{K_{out} \cap B_\delta(\mathbf{r})} \mathbf{v}(\mathbf{x}) \cdot \int_{\Gamma_K \cap B_\delta(\mathbf{x})} 2\Lambda \cdot \\
&\quad (\nabla \mathbf{u}_{\Gamma_K}(\mathbf{s}) - \nabla \mathbf{u}_K(\mathbf{x})) \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\mathbf{x} \\
&\quad + \int_{\mathbf{e}} \mathbf{v}(\mathbf{x}_{\mathbf{e}}) \cdot \int_{-\delta}^0 \int_{B_\delta(\mathbf{x}) \cap \Gamma_K} 2\Lambda \cdot \\
(4.25) \quad &\quad (\nabla \mathbf{u}_{\Gamma_K}(\mathbf{x}_{\mathbf{e}}) - \nabla \mathbf{u}_K(\mathbf{x}_{\mathbf{e}})) \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\xi d\vec{\eta} + o(1),
\end{aligned}$$

where $\mathbf{x}_{\mathbf{e}} = (\xi, 0)$. For the K' part, we change the order of integration and obtain

$$\begin{aligned}
S'_2 &= \int_{K'} \mathbf{v}(\mathbf{x}) \cdot \int_{\Gamma_{K'}} 2\Lambda \cdot (\nabla \mathbf{u}_{\Gamma_{K'}}(\mathbf{s}) - \nabla \mathbf{u}_{K'}(\mathbf{x})) \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\mathbf{x} \\
&= \sum_{\mathbf{r} \in \text{edge}(K')/\{\mathbf{e}\}} \int_{K'_{out} \cap B_\delta(\mathbf{r})} \mathbf{v}(\mathbf{x}) \cdot \int_{\Gamma_{K'} \cap B_\delta(\mathbf{x})} 2\Lambda \cdot \\
&\quad (\nabla \mathbf{u}_{\Gamma_{K'}}(\mathbf{s}) - \nabla \mathbf{u}_{K'}(\mathbf{x})) \cdot (\mathbf{x}' - \mathbf{s}) d\mathbf{x}' d\mathbf{x} \\
&\quad + \int_{\mathbf{e}} \mathbf{v}(\mathbf{x}_{\mathbf{e}}) \cdot \int_{-\delta}^0 \int_{B_\delta(\mathbf{x}) \cap \Gamma_K} 2\Lambda \cdot \\
(4.26) \quad &\quad (\nabla \mathbf{u}_{\Gamma_K}(\mathbf{x}_{\mathbf{e}}) - \nabla \mathbf{u}_K(\mathbf{x}_{\mathbf{e}})) \cdot (\mathbf{s} - \mathbf{x}) d\mathbf{x}' d\xi d\vec{\eta} + o(1).
\end{aligned}$$

Adding up (4.25) and (4.26), the resulting integral on edge (face) $\mathbf{e}$ can be presented as

$$(4.27) \quad \int_{\mathbf{e}} \mathbf{v}(\mathbf{x}_{\mathbf{e}}) \cdot \int_{-\delta}^0 \int_{SC(\delta, \delta + \xi)} \frac{2\beta}{|\mathbf{z}|^2} \mathbf{z} \otimes \mathbf{z} \nabla \tilde{\mathbf{u}}(\mathbf{x}_{\mathbf{e}}) \mathbf{z} d\mathbf{z} d\xi d\vec{\eta} + o(1),$$

where $\mathbf{z} = \mathbf{x}' - \mathbf{x}$, $\nabla \tilde{\mathbf{u}}(\cdot) = \nabla \mathbf{u}_{\Gamma_K}(\cdot) - \nabla \mathbf{u}_K(\cdot)$. Notice the contributions from S_2 and S'_2 to (4.27) are the same as $\delta \rightarrow 0$.

The i -th component of $\mathbf{z} \otimes \mathbf{z} \nabla \tilde{\mathbf{u}}(\mathbf{x}_e) \mathbf{z}$ can be computed as

$$(4.28) \quad (\mathbf{z} \otimes \mathbf{z} \nabla \tilde{\mathbf{u}}(\mathbf{x}_e) \mathbf{z})_i = \sum_{j,k=1}^d z_i z_j z_k \tilde{u}_{j,k}(\mathbf{x}_e),$$

where $\tilde{u}_{j,k}$ denotes the derivative of $u_j|_{\Gamma_K} - u_j|_K$ along x_k direction. Since the outward normal of K on $\mathbf{e}$ is $(1, 0, 0, \dots, 0)^T$, by the symmetry of the integral, when $i = 1$ we have

$$(4.29) \quad \begin{aligned} & \int_{-\delta}^0 \int_{B_\delta(\mathbf{x}) \cap \Gamma_K} \frac{2\beta}{|\mathbf{z}|^2} \left(\sum_{j,k=1}^d z_i z_j z_k \tilde{u}_{j,k}(\mathbf{x}_e) \right) d\mathbf{z} d\eta \\ &= \int_{-\delta}^0 \int_{B_\delta(\mathbf{x}) \cap \Gamma_K} \frac{2\beta}{|\mathbf{z}|^2} \left(\sum_{j=1}^d z_1 z_j^2 \tilde{u}_{j,j}(\mathbf{x}_e) \right) d\mathbf{z} d\eta \\ &= \int_{B_\delta(\mathbf{x}_e) \cap \Gamma_K} \int_{-z_1}^0 \frac{2\beta}{|\mathbf{z}|^2} \left(\sum_{j=1}^d z_1 z_j^2 \tilde{u}_{j,j}(\mathbf{x}_e) \right) d\eta d\mathbf{z} \\ &= \int_{B_\delta(0)} \frac{\beta}{|\mathbf{z}|^2} \left(\sum_{j=1}^d z_1^2 z_j^2 \tilde{u}_{j,j}(\mathbf{x}_e) \right) d\mathbf{z} \\ &= \left(\int_{B_\delta(0)} \frac{\beta}{|\mathbf{z}|^2} z_1^2 z_2^2 d\mathbf{z} \right) (3\tilde{u}_{1,1}(\mathbf{x}_e) + \sum_{j=2}^d \tilde{u}_{j,j}(\mathbf{x}_e)). \end{aligned}$$

Similarly, when $i > 1$ we get

$$(4.30) \quad \begin{aligned} & \int_{-\delta}^0 \int_{B_\delta(\mathbf{x}) \cap \Gamma_K} \frac{2\beta}{|\mathbf{z}|^2} \left(\sum_{j,k=1}^d z_i z_j z_k \tilde{u}_{j,k}(\mathbf{x}_e) \right) d\mathbf{z} d\eta \\ &= \int_{-\delta}^0 \int_{B_\delta(\mathbf{x}) \cap \Gamma_K} \frac{2\beta}{|\mathbf{z}|^2} (z_1 z_i^2 (\tilde{u}_{1,i}(\mathbf{x}_e) + \tilde{u}_{i,1}(\mathbf{x}_e))) d\mathbf{z} d\eta \\ &= \int_{B_\delta(0)} \frac{\beta}{|\mathbf{z}|^2} (z_1^2 z_i^2 (\tilde{u}_{1,i}(\mathbf{x}_e) + \tilde{u}_{i,1}(\mathbf{x}_e))) d\mathbf{z} \\ &= \left(\int_{B_\delta(0)} \frac{\beta}{|\mathbf{z}|^2} z_1^2 z_2^2 d\mathbf{z} \right) (\tilde{u}_{1,i}(\mathbf{x}_e) + \tilde{u}_{i,1}(\mathbf{x}_e)). \end{aligned}$$

Combining (4.29) and (4.30), we get that the expression in (4.27) becomes

$$\int_{\mathbf{e}} \mathbf{v}(\mathbf{x}_e) \cdot \mu_\delta (\nabla \tilde{\mathbf{u}}(\mathbf{x}_e) + \nabla \tilde{\mathbf{u}}^T(\mathbf{x}_e) + (\nabla \cdot \tilde{\mathbf{u}}(\mathbf{x}_e)) \mathbf{I}) \cdot \tilde{\mathbf{n}}_e d\xi + o(1),$$

where

$$\mu_\delta = \int_{B_\delta(0)} \frac{\beta}{|\mathbf{z}|^2} z_1^2 z_2^2 d\mathbf{z}.$$

So

$$(4.31) \quad S_2 \rightarrow -\frac{1}{2} \int_{\partial K} \mathbf{v}(\nu) \cdot \mu \llbracket \nabla \mathbf{u}(\nu) + \nabla^T \mathbf{u}(\nu) + (\nabla \cdot \mathbf{u}(\nu)) \mathbf{I} \rrbracket_{\mathbf{r}} d\nu,$$

where $\mu = \lim_{\delta \rightarrow 0} \mu_\delta$.

Finally, combining (4.16), (4.17), (4.21), (4.23) and (4.31), we have

Theorem 4.3. *Under the assumption (4.3), with $\mathbf{u}$ and $\mathbf{v}$ satisfying (4.15). For any element K , as $\delta \rightarrow 0$,*

$$\begin{aligned}
 & - \int_K \mathbf{v}(\mathbf{x}) \cdot \mathcal{G}(\beta \mathcal{G}^*(\mathbf{u})) d\mathbf{x}' d\mathbf{x} \\
 \rightarrow & \int_K \mathbf{v}(\mathbf{x}) \cdot (\mu(\nabla \cdot (\nabla \mathbf{u} + \nabla^T \mathbf{u}) + \nabla(\nabla \cdot \mathbf{u}))) d\mathbf{x} \\
 (4.32) \quad & - \frac{1}{2} \int_{\partial K} \mathbf{v}(\nu) \cdot \mu[\nabla \mathbf{u}(\nu) + \nabla^T \mathbf{u}(\nu) + (\nabla \cdot \mathbf{u}(\nu))\mathbf{I}]_{\mathbf{r}} d\nu.
 \end{aligned}$$

We therefore have the corollary

Corollary 4.4. *Under assumption (4.3), with $\mathbf{u}^h$ and $\mathbf{v}$ satisfy (4.15). For any element K , as $\delta \rightarrow 0$,*

$$(4.33) \quad \int_K \mathbf{v}(\mathbf{x}) \cdot \mathbf{R}^h d\mathbf{x}' d\mathbf{x} \rightarrow \int_K \mathbf{v}(\mathbf{x}) \cdot \mathbf{r}_*^h d\mathbf{x} - \frac{1}{2} \int_{\partial K} \mathbf{v}(\nu) \cdot \mathbf{R}_*^h d\nu.$$

Thus we can see that the a posteriori error estimation for the PD bond-based model is connected to its local counterpart as well. Summing up (4.33) for all the K 's implies the local limit of PD operator being the Navier operator (with Poisson ratio 1/4), as stated in [21].

5. NUMERICAL EXPERIMENTS

In this section, we compute the finite element approximations and the a posteriori error approximations for several nonlocal diffusion equations to validate our theoretical analysis.

Example 5.1. We consider the following 1D problem,

$$(5.1) \quad \begin{cases} -\frac{1}{\delta^3} \int_{x-\delta}^{x+\delta} (u(x') - u(x)) dx' = f(x), & \forall x \in [0, 1], \\ u(x) = g(x), & \forall x \in [-\delta, 0] \cup [1, 1 + \delta]. \end{cases}$$

We choose the exact solution to be

$$u(x) = -x^3 + x^2,$$

therefore

$$f(x) = 2x - \frac{2}{3}.$$

Also, we set $\delta = 2, 0.2, 0.02$, respectively, and restriction condition $g(x)$ can be determined by $u(x)$ and δ accordingly. Then we can compute $M(\delta)$ as defined in (2.9) by

$$M(\delta) = \int_{-\delta}^{\delta} \frac{1}{\delta^3} dz = \frac{2}{\delta^2},$$

so $M(\delta) = 0.5, 50, 5000$ for different δ .

We use continuous Galerkin method to solve problem (5.1) with continuous piecewise linear basis functions, one can refer to [19] for more implementation details. For each δ , we divide the interval $[0, 1]$ into N elements, where $N = 25, 50, 100, 200, 400$, respectively. The numerical results can be found in Table

$\delta = 2$								
N	$\ e^h\ _{L^2}$	CR	$\ e^h\ _E$	CR	η_δ^h	CR	$\eta_\delta^h/\ e^h\ _{L^2}$	$\eta_\delta^h/\ e^h\ _E$
25	2.9212e-04	-	8.4472e-05	-	8.0850e-05	-	0.277	0.957
50	7.3030e-05	2.00	2.1118e-05	2.00	2.0213e-05	2.00	0.277	0.957
100	1.8257e-05	2.00	5.2795e-06	2.00	5.0531e-06	2.00	0.277	0.957
200	4.5644e-06	2.00	1.3199e-06	2.00	1.2633e-06	2.00	0.277	0.957
400	1.1411e-06	2.00	3.2997e-07	2.00	3.1582e-07	2.00	0.277	0.957
$\delta = 0.2$								
N	$\ e^h\ _{L^2}$	CR	$\ e^h\ _E$	CR	η_δ^h	CR	$\eta_\delta^h/\ e^h\ _{L^2}$	$\eta_\delta^h/\ e^h\ _E$
25	2.9212e-04	-	8.4472e-04	-	8.0850e-04	-	2.768	0.957
50	7.3030e-05	2.00	2.1118e-04	2.00	2.0213e-04	2.00	2.768	0.957
100	1.8257e-05	2.00	5.2795e-05	2.00	5.0531e-05	2.00	2.768	0.957
200	4.5644e-06	2.00	1.3199e-05	2.00	1.2633e-05	2.00	2.768	0.957
400	1.1411e-06	2.00	3.2997e-06	2.00	3.1582e-06	2.00	2.768	0.957
$\delta = 0.02$								
N	$\ e^h\ _{L^2}$	CR	$\ e^h\ _E$	CR	η_δ^h	CR	$\eta_\delta^h/\ e^h\ _{L^2}$	$\eta_\delta^h/\ e^h\ _E$
25	2.9212e-04	-	8.4472e-03	-	8.0850e-03	-	27.677	0.957
50	7.3030e-05	2.00	2.1118e-03	2.00	2.0213e-03	2.00	27.677	0.957
100	1.8257e-05	2.00	5.2795e-04	2.00	5.0531e-04	2.00	27.677	0.957
200	4.5644e-06	2.00	1.3199e-04	2.00	1.2633e-04	2.00	27.677	0.957
400	1.1411e-06	2.00	3.2997e-05	2.00	3.1582e-05	2.00	27.677	0.957

TABLE 1. Numerical results with different δ in Example 5.1.

1, where we list the error in L^2 norm, in energy norm, and a posteriori error estimator. The convergence rates for each error are computed as well as the ratio between η_δ^h and $\|e^h\|_E$ to examine reliability and efficiency of the estimator. We can observe that our a posteriori error estimator approximates $\|e^h\|_E$ robustly, the ratio between the two is very stable for each δ , so our estimator is both reliable and efficient. This numerical experiment is consistent with Theorem 3.4. One can also observe that the ratio between η_δ^h and $\|e^h\|_{L^2}$ is also stable for fixed δ , while the ratio is proportional to $\sqrt{M(\delta)}$, which is not surprising given the loss of equivalence between the energy norm and the L^2 norm in the local limit.

Example 5.2. We use the same problem as in Example 5.1 but with a discontinuous exact solution

$$(5.2) \quad u(x) = \begin{cases} x, & \forall x < 0.5, \\ x^2, & \text{otherwise.} \end{cases}$$

Therefore the right side term can be found as

$$(5.3) \quad f(x) = \begin{cases} 0, & \forall x < 0.5 - \delta, \\ -\frac{1}{\delta^3} \left(\frac{(x+\delta)^3}{3} - \frac{(x+\delta)^2}{2} + \frac{1}{12} \right), & \forall 0.5 - \delta \leq x < 0.5, \\ -\frac{1}{\delta^3} \left(\frac{(x-\delta)^3}{3} - \frac{(x-\delta)^2}{2} + \frac{1}{12} \right) - \frac{2}{3}, & \forall 0.5 \leq x < 0.5 + \delta, \\ -\frac{2}{3}, & \text{otherwise.} \end{cases}$$

We study the numerical solution with continuous piecewise linear basis functions and $\delta = 2, 0.2, 0.02$.

$\delta = 2$								
N	$\ e^h\ _{L^2}$	CR	$\ e^h\ _E$	CR	η_δ^h	CR	$\eta_\delta^h/\ e^h\ _{L^2}$	$\eta_\delta^h/\ e^h\ _E$
25	1.3808e-02	-	9.7711e-03	-	9.7702e-03	-	0.708	1.000
50	1.3436e-02	0.04	9.3357e-03	0.07	9.3358e-03	0.07	0.695	1.000
100	9.4986e-03	0.50	6.6000e-03	0.50	6.6000e-03	0.50	0.695	1.000
200	6.7162e-03	0.50	4.6667e-03	0.50	4.6667e-03	0.50	0.695	1.000
400	4.7490e-03	0.50	3.3003e-03	0.50	3.3003e-03	0.50	0.695	1.000
$\delta = 0.2$								
N	$\ e^h\ _{L^2}$	CR	$\ e^h\ _E$	CR	η_δ^h	CR	$\eta_\delta^h/\ e^h\ _{L^2}$	$\eta_\delta^h/\ e^h\ _E$
25	1.9657e-02	-	1.0062e-01	-	9.7778e-02	-	4.974	0.972
50	1.3437e-02	0.55	9.3350e-02	0.11	9.3347e-02	0.07	6.947	1.000
100	9.4988e-03	0.50	6.5999e-02	0.50	6.5998e-02	0.50	6.948	1.000
200	6.7162e-03	0.50	4.6666e-02	0.50	4.6666e-02	0.50	6.948	1.000
400	4.7490e-03	0.50	3.2998e-02	0.50	3.2998e-02	0.50	6.948	1.000
$\delta = 0.02$								
N	$\ e^h\ _{L^2}$	CR	$\ e^h\ _E$	CR	η_δ^h	CR	$\eta_\delta^h/\ e^h\ _{L^2}$	$\eta_\delta^h/\ e^h\ _E$
25	1.9430e-02	-	1.3555e+00	-	1.3197e+00	-	67.924	0.974
50	1.3574e-02	0.52	9.6055e-01	0.50	9.8866e-01	0.42	72.837	1.029
100	9.5121e-03	0.51	6.5267e-01	0.56	6.4654e-01	0.61	67.970	0.991
200	6.7193e-03	0.50	4.6610e-01	0.49	4.6571e-01	0.47	69.309	0.999
400	4.7496e-03	0.50	3.2994e-01	0.50	3.2992e-01	0.50	69.462	1.000

TABLE 2. Numerical results with different δ in Example 5.2.

The interval $[0, 1]$ is divided into $N = 25, 50, 100, 200, 400$ subintervals respectively and numerical results can be found in Table 2. The convergence rate of $\|e^h\|_E$ is found to be about $1/2$ (consistent with the numerical experiments in [19]). The a posteriori error estimator has the same convergence rate, and the ratio between the two remains almost a constant along with the increase of N , which indicates the reliability and efficiency numerically. It again reveals that the ratios $\eta_\delta^h/\|e^h\|_{L^2}$ are proportional to δ^{-1} , similar to the previous examples.

Since the exact solution is discontinuous, our error estimator should serve to capture the discontinuity (at 0.5). Figure 6 and 7 show distributions of $\log_{10}(\|e^h\|_E)$ and $\log_{10}(\eta^h)$ for all the grids over $[0, 1]$, with different δ and N . It can be observed that our error estimator approximates the exact error very well, and the larger N is, the better the estimator gets in capturing the discontinuity.

Example 5.3. In this example, we resolve Example 5.2 with discontinuous piecewise linear basis functions and $\delta = 2, 0.2, 0.02$. Computational results are listed in Table 3. As discussed in [19], we get 2nd order convergence for L^2 norm of error with discontinuous piecewise linear basis, and we also get 2nd order convergence for both energy norm of error and posteriori error estimator. Notice that, for Examples 5.2 and 5.3, the discontinuity point is on one of the grid points.

Example 5.4. We use the same problem as in Example 5.1 but with a discontinuous exact solution whose discontinuity is inside one of the elements (for $N = 25, 50, 100, 200, 400$), defined as following

$$(5.4) \quad u(x) = \begin{cases} x, & \forall x < 0.503, \\ x^2, & \text{otherwise.} \end{cases}$$

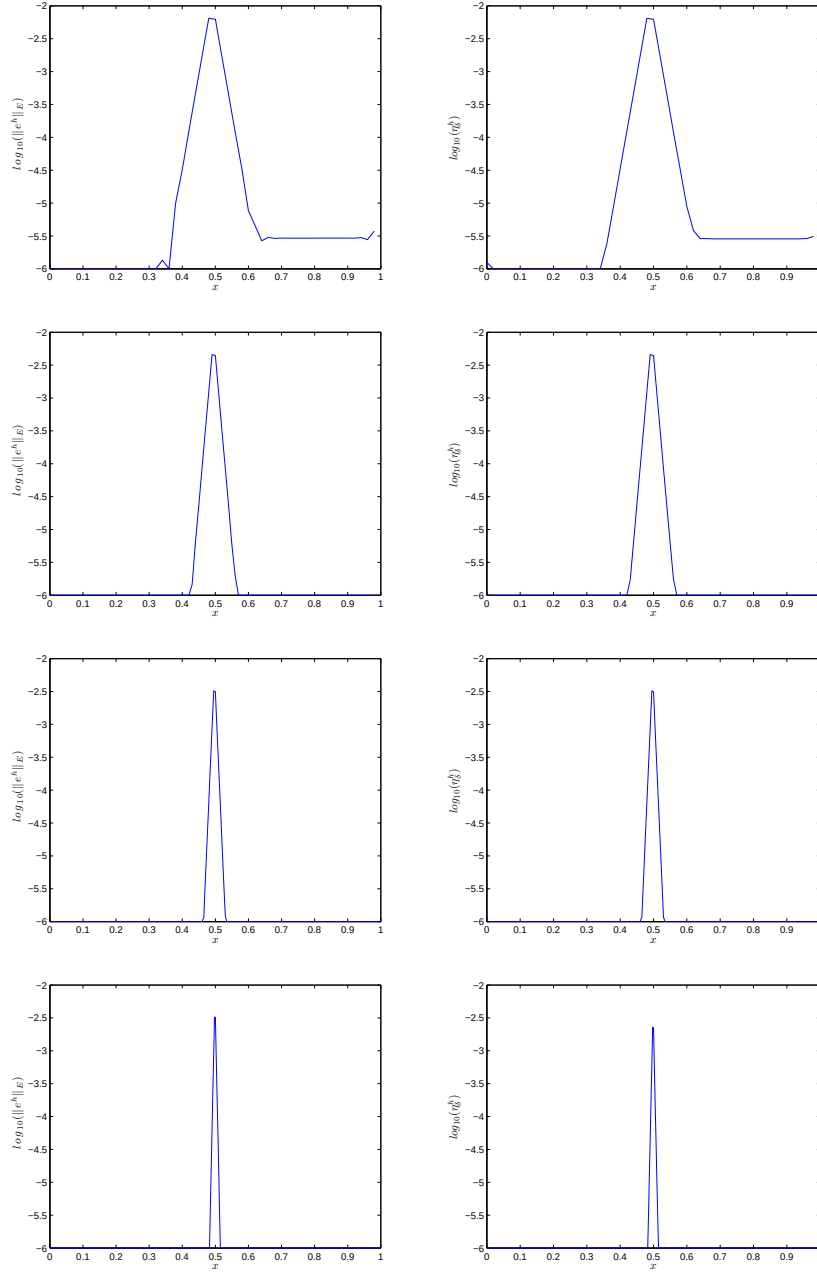


FIGURE 6. Error distributions of exact error (left) and estimator (right) for Example 5.2 with $\delta = 2$, $N = 50$, $N = 100$, $N = 200$ and $N = 400$ (from top to bottom).

Therefore the right side term can be found as

$$(5.5) \quad f(x) = \begin{cases} 0, & \forall x < 0.503 - \delta, \\ -\frac{1}{\delta^3} \left(\frac{(x + \delta)^3}{3} - \frac{(x + \delta)^2}{2} + \frac{1}{12} \right), & \forall 0.503 - \delta \leq x < 0.503, \\ -\frac{1}{\delta^3} \left(\frac{(x - \delta)^3}{3} - \frac{(x - \delta)^2}{2} + \frac{1}{12} \right) - \frac{2}{3}, & \forall 0.503 \leq x < 0.503 + \delta, \\ -\frac{2}{3}, & \text{otherwise.} \end{cases}$$

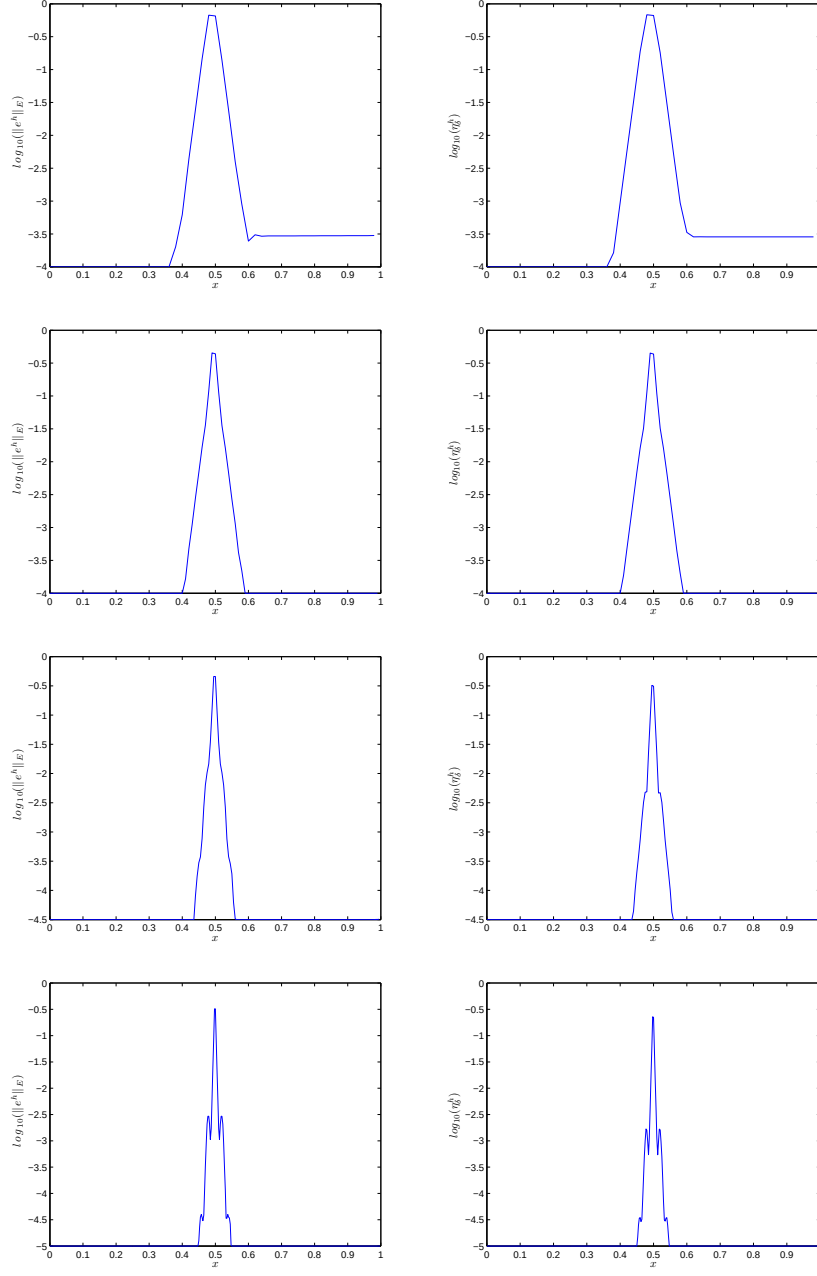


FIGURE 7. Error distributions of exact error (left) and estimator (right) for Example 5.2 with $\delta = 0.02$, $N = 50$, $N = 100$, $N = 200$ and $N = 400$ (from top to bottom).

We study the numerical solution with discontinuous piecewise linear basis functions and $\delta = 2, 0.2, 0.02$. Computational results are listed in Table 4, one can observe

$\delta = 2$								
N	$\ e^h\ _{L^2}$	CR	$\ e^h\ _E$	CR	η_δ^h	CR	$\eta_\delta^h / \ e^h\ _{L^2}$	$\eta_\delta^h / \ e^h\ _E$
25	1.2848e-02	-	9.0822e-03	-	9.0800e-03	-	0.707	1.000
50	3.9030e-05	8.36	1.5046e-05	9.24	1.5011e-05	9.24	0.385	0.998
100	9.8456e-06	1.99	3.7442e-06	2.01	3.7398e-06	2.00	0.380	0.999
200	2.4863e-06	1.99	9.3605e-07	2.00	9.3496e-07	2.00	0.376	0.999
400	6.2784e-07	1.99	2.3401e-07	2.00	2.3374e-07	2.00	0.372	0.999
$\delta = 0.2$								
N	$\ e^h\ _{L^2}$	CR	$\ e^h\ _E$	CR	η_δ^h	CR	$\eta_\delta^h / \ e^h\ _{L^2}$	$\eta_\delta^h / \ e^h\ _E$
25	1.2830e-02	-	9.0735e-02	-	9.0765e-02	-	7.075	1.000
50	4.1913e-05	8.26	1.4933e-04	9.25	1.4910e-04	9.25	3.557	0.998
100	1.0611e-05	1.98	3.7301e-05	2.00	3.7272e-05	2.00	3.513	0.999
200	2.6690e-06	1.99	9.3211e-06	2.00	9.3175e-06	2.00	3.491	1.000
400	6.7150e-07	1.99	2.3298e-06	2.00	2.3293e-06	2.00	3.469	1.000
$\delta = 0.02$								
N	$\ e^h\ _{L^2}$	CR	$\ e^h\ _E$	CR	η_δ^h	CR	$\eta_\delta^h / \ e^h\ _{L^2}$	$\eta_\delta^h / \ e^h\ _E$
25	1.2818e-02	-	9.0556e-01	-	9.0741e-01	-	70.792	1.002
50	4.2893e-05	8.22	1.4908e-03	9.25	1.4906e-03	9.25	34.752	1.000
100	1.0831e-05	1.99	3.7269e-04	2.00	3.7265e-04	2.00	34.404	1.000
200	2.7206e-06	1.99	9.3173e-05	2.00	9.3168e-05	2.00	34.245	1.000
400	6.8441e-07	1.99	2.3293e-05	2.00	2.3292e-05	2.00	34.032	1.000

TABLE 3. Numerical results with different δ in Example 5.3.

that the convergence rates for the norms and estimator decrease to 0.5, this is consistent with results in [19]. Also our estimator approximates the exact error robustly.

Example 5.5. In the last example, we take a singular kernel function and consider the following problem

$$(5.6) \quad \begin{cases} -\frac{1}{\delta^2 \sqrt{\delta}} \int_{x-\delta}^{x+\delta} \frac{u(x') - u(x)}{\sqrt{|x' - x|}} dx' = f(x), & \forall x \in [0, 1], \\ u(x) = g(x), & \forall x \in [-\delta, 0] \cup [1, 1 + \delta]. \end{cases}$$

We set the exact solution

$$u(x) = x^2,$$

and therefore the right side term

$$f(x) = -\frac{4}{5}.$$

The boundary term $g(x)$ is decided from the equation (5.6). We solve the problem with $\delta = 2, 0.2$ respectively.

As in previous examples, we use continuous galerkin method with piecewise linear basis functions. Again, for fixed δ , we divide the interval $[0, 1]$ into $N = 25, 50, 100, 200, 400$ subintervals, respectively. Numerical results are shown in Table 5. We again can observe that the error estimator has the same convergence rate with $\|e^h\|_E$, and the ratio between the two is very stable, where the efficiency can be observed. For the kernel function used in this particular example, we also have

$$\sqrt{M(\delta)} \sim \sqrt{\frac{1}{\delta^2 \sqrt{\delta}} \int_{-\delta}^{\delta} \frac{1}{\sqrt{|z|}} dz} \sim \frac{1}{\delta},$$

$\delta = 2$								
N	$\ e^h\ _{L^2}$	CR	$\ e^h\ _E$	CR	η_δ^h	CR	$\eta_\delta^h/\ e^h\ _{L^2}$	$\eta_\delta^h/\ e^h\ _E$
25	1.2420e-02	-	8.8631e-03	-	8.8622e-03	-	0.714	1.000
50	9.7694e-03	0.35	6.8992e-03	0.36	6.8999e-03	0.36	0.706	1.000
100	6.8959e-03	0.50	4.8251e-03	0.52	4.8249e-03	0.52	0.700	1.000
200	5.0843e-03	0.44	3.4118e-03	0.50	3.4123e-03	0.50	0.671	1.000
400	3.3486e-03	0.60	2.4135e-03	0.50	2.4131e-03	0.50	0.721	1.000
$\delta = 0.2$								
N	$\ e^h\ _{L^2}$	CR	$\ e^h\ _E$	CR	η_δ^h	CR	$\eta_\delta^h/\ e^h\ _{L^2}$	$\eta_\delta^h/\ e^h\ _E$
25	1.2535e-02	-	8.8622e-02	-	8.8622e-02	-	7.070	1.000
50	9.7589e-03	0.36	6.9001e-02	0.36	6.8999e-02	0.36	7.070	1.000
100	6.8236e-03	0.52	4.8250e-02	0.52	4.8251e-02	0.52	7.071	1.000
200	4.8258e-03	0.50	3.4123e-02	0.50	3.4123e-02	0.50	7.071	1.000
400	3.4153e-03	0.50	2.4132e-02	0.50	2.4132e-02	0.50	7.066	1.000
$\delta = 0.02$								
N	$\ e^h\ _{L^2}$	CR	$\ e^h\ _E$	CR	η_δ^h	CR	$\eta_\delta^h/\ e^h\ _{L^2}$	$\eta_\delta^h/\ e^h\ _E$
25	1.2395e-02	-	8.8629e-01	-	8.8764e-01	-	71.614	1.002
50	9.6166e-03	0.37	6.9009e-01	0.36	6.9029e-01	0.36	71.780	1.000
100	6.8725e-03	0.48	4.8248e-01	0.52	4.8271e-01	0.52	70.238	1.000
200	4.8080e-03	0.52	3.4125e-01	0.50	3.4282e-01	0.49	71.301	1.005
400	3.3957e-03	0.50	2.4121e-01	0.50	2.4052e-01	0.51	70.831	0.997

TABLE 4. Numerical results with different δ in Example 5.4.

$\delta = 2$								
N	$\ e^h\ _{L^2}$	CR	$\ e^h\ _E$	CR	η_δ^h	CR	$\eta_\delta^h/\ e^h\ _{L^2}$	$\eta_\delta^h/\ e^h\ _E$
25	2.9256e-04	-	1.1747e-04	-	1.1624e-04	-	0.397	0.990
50	7.3109e-05	2.00	2.9501e-05	1.99	2.9625e-05	1.97	0.405	1.004
100	1.8272e-05	2.00	7.3983e-06	2.00	7.5890e-06	1.96	0.415	1.026
200	4.5669e-06	2.00	1.8536e-06	2.00	1.9697e-06	1.95	0.431	1.063
400	1.1415e-06	2.00	4.6339e-07	2.00	4.9116e-07	2.00	0.430	1.060
$\delta = 0.2$								
N	$\ e^h\ _{L^2}$	CR	$\ e^h\ _E$	CR	η_δ^h	CR	$\eta_\delta^h/\ e^h\ _{L^2}$	$\eta_\delta^h/\ e^h\ _E$
25	3.1313e-04	-	1.1341e-03	-	1.1751e-03	-	3.753	1.036
50	7.6830e-05	2.03	2.8800e-04	1.98	2.9674e-04	1.99	3.862	1.030
100	1.8937e-05	2.02	7.2764e-05	1.98	7.5125e-05	1.98	3.967	1.032
200	4.6852e-06	2.02	1.8323e-05	1.99	1.8906e-05	1.99	4.035	1.032
400	1.1625e-06	2.01	4.6038e-06	1.99	4.7426e-06	2.00	4.080	1.030

TABLE 5. Numerical results with different δ in Example 5.5.

which again explains the growth of $\eta_\delta^h/\|e^h\|_{L^2}$.

6. CONCLUSION

In this work, a rigorous mathematical framework is established for the a posteriori error analysis of the finite element solutions for nonlocal diffusion and nonlocal peridynamic equations. The estimators are carefully derived and the reliability and efficiency are proven. The numerical experiments are consistent with our theoretical conclusion. Meanwhile, the weak convergence of the nonlocal estimator to its local analogue is demonstrated for nonlocal diffusion equation and peridynamic model problem, drawing interesting connections to the limit case for which a lot of studies can be found in the literature. The nonlocal nature of the problem also

offers some sharp contrasts with the local counterpart as explored in our analysis. While the analysis is done for problems with an operator-induced norm being equivalent to L^2 norm, due to our choice of kernel function, it is possible to generalize such a framework of a posteriori error analysis for problems involving more general kernel functions. This will be pursued in our future works, along with the development of the adaptive convergence analysis and practical implementation for nonlocal models.

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